

Group Project Report

Estimating the Financial Impact of E-Commerce Returns: A Transaction-Level Expected Return Loss (ERL) Framework

MSc Data Science and Its Application

~~CONFIDENTIAL~~ Group Project

Group Number: 4

Names: Afrida Nawer Prami, Dang Nguyet Anh Nguyen, Julia Legner, Sawsan Ibrahim

Supervisor: ~~CONFIDENTIAL~~

Submission Date: 17/06/2026

Word count: 7.386

Dates of group meetings: Meeting 1: 04/05/2026, Meeting 2: 11/05/2026, Meeting 3: 20/05/2026, Meeting 4: 25/05/2026, Meeting 5: 28/05/2026, Meeting 6: 01/06/2026, Meeting 7: 08/06/2026

i. Abstract

E-commerce returns cost online fashion retailers far more than lost revenue alone. Each returned transaction triggers a chain of costs: return shipping, receiving and inspection, restocking or disposal, and in many cases a markdown on the resale value of the item. These costs vary significantly depending on why the item was returned. A defective item requires inspection and potentially disposal. A size issue may be restocked. A changed mind may require a markdown. Existing research predicts whether a return will happen but does not estimate what it will cost. *This gap means retailers have no principled basis for deciding which returns are worth trying to prevent and which are not.*

This project develops the Expected Return Loss (ERL) framework to fill that gap. The framework combines return probability, return reason classification, and reason-specific processing cost into a single financial risk score per transaction. It was implemented across four connected sub-topics on TheLook, a fully synthetic e-commerce dataset published by Google for research purposes, comprising 85,187 fictional transactions joined to an Amazon Apparel review corpus. Sub-Topic 1(ST1) trained two classifiers to predict return probability from purchase-time features. Neither reached the required validation threshold of AUC-ROC 0.80, so a scenario-based score anchored to category return averages was used in its place. Sub-Topic 2 (ST2) classified return reasons from customer review text using Snorkel weak supervision and a DistilBERT transformer classifier, achieving a macro F1 of 0.737 across five reason categories. Three categories performed reliably; two did not due to structural scarcity of training signal.

Sub-Topic 3(ST3) combined the ST1 and ST2 outputs with a literature-grounded cost matrix to produce the ERL score per transaction. ERL-based ranking shifted 95% of transactions by more than 1,000 positions compared with return-probability ranking, raising the Gini coefficient from 0.106 to 0.160. Sub-Topic 4 evaluated return-prevention interventions across two modelling approaches. Broad transaction-level rollout produced a total net loss of £125,110. Selective targeting of the 31 highest-risk profitable transactions produced a positive return of £3.10. A category-level model produced positive net benefits across multiple product groups, with Jeans achieving the strongest result at +£2,184.75. The project demonstrates that financial risk and return likelihood are structurally different quantities, and that the ERL framework: as the first integration of return prediction, reason classification, and cost estimation at transaction level, provides a more financially discriminating basis for intervention decisions than probability-based systems alone. The framework is presented as a validated prototype and a foundation for retailer-specific deployment.

ii. Acknowledgement

The authors would like to thank Dr Ebrahim Patel for his guidance and support throughout this project as our supervisor, and Dr Stef Garasto for his leadership as Module Leader of the MSc Data Science and Its Applications programme at the University of Greenwich.

We would also like to acknowledge one another as a team: Afrida Nawer Prami, Dang Nguyet Anh Nguyen, Julia Legner, and Sawsan Hussain Saleh Ibrahim, for the collaborative effort, mutual support, and commitment that made this project possible despite the challenges of balancing group work across a demanding postgraduate programme.

Finally, we extend our gratitude to our families and loved ones for their patience and encouragement throughout our studies. This project marks the conclusion of our MSc journey and reflects the dedication each of us has brought to it.

iii. TABLE OF CONTENTS

Contents

i.	Abstract	i
ii.	Acknowledgement	ii
iii.	TABLE OF CONTENTS	iii
iv.	Framework Overview	vi
v.	List of Figures	vii
vi.	List of Tables	viii
vii.	List of Abbreviations	ix
1.	Introduction.....	1
2.	LITERATURE REVIEW	3
2.1.	Sub-topic 1: Return Probability Modelling.....	3
2.2.	Sub-topic 2: Return Reason Classification.....	4
2.3.	Sub-topic 3: ERL Calculation	4
2.4.	Sub-topic 4: Intervention Evaluation.....	5
2.5.	Project's Purpose	6
2.6.	Ethical and Fairness Considerations	6
3.	Legal, Social, Ethical and Professional Issues	7
4.	Methods: Framework Overview and Pipeline Design.....	9
3.	Sub-Topic 1: Return Probability Modelling	11
3.1.	Methods and Data	11
3.2.	Implementation.....	12
3.3.	Results	13
3.4.	Discussion and Limitations.....	16
4.	Sub-Topic 2: Return Reason Classification	18
4.1.	Methods and Data	18
4.2.	Implementation.....	19

4.3. Results	20
4.4. Discussion and Limitations.....	21
5. Sub-Topic 3: ERL Construction	23
5.1. Methods and Data	23
5.2. Implementation.....	24
5.3. Results	24
5.4. Discussion and Limitations.....	27
6. Sub-Topic 4: Intervention Evaluation	30
6.1. Methods.....	30
Approach A: transaction-level intervention model	30
Approach B: category-level strategic model	31
6.2. Results	32
Approach A: transaction-level results	32
Approach B: category-level results	32
Sensitivity analysis.....	34
6.3. Discussion and Limitations.....	35
7. Conclusion and Discussion	37
7.1. Cross-Pipeline Integration and Combined Findings.....	37
7.2. Overall Conclusion	38
8. Author Contributions	39
9. References.....	40
10. Appendix.....	43
Appendix A: ST1 Deployment Validation Run	43
A1 — Category Return-Rate Diagnostic.....	43
A2 — Validation Model Comparison and Threshold Decision	43
A3 — Final Test Assessment.....	44
Appendix B: ST2: Return Reason Classification Pipeline and	45

Full P(reason product_type) Matrix	45
Overview	45
End-to-end pipeline	45
Sample data through the pipeline (same 3 reviews)	45
Phase 3: Manual annotation validation	47
Phase 4: DistilBERT fine-tuning	47
Final output: P(reason product_category)	47
Final output: Full P(reason product_type) Matrix	48
Appendix C: Code and Reproducibility	49
Appendix D: ST3 Risk Tier Reason Decomposition.....	50
Appendix E: ERL Formula Worked Examples– ST3	50
Appendix F: ST4 Worked Examples.....	51
11. Declaration of AI Use	53

iv. Framework Overview

Report Section	ST1: Return Probability	ST2: Reason Classification	ST3: ERL Construction	ST4: Intervention Evaluation
Research question	Can we predict which purchases will be returned using information available at the time of purchase?	Can we identify why a return happens by analysing the text customers write in online reviews?	Can we combine the return prediction and reason classification into a single financial risk score per transaction?	Which actions are worth taking to reduce return costs, and is it better to act on individual transactions or on product groups?
Approach used	Two machine learning classifiers trained on 85,187 purchase records, then tested against a threshold of AUC 0.80	Text classification pipeline using 795,000 Amazon apparel reviews, weak supervision labelling, and a DistilBERT transformer model	A cost formula that multiplies return probability by reason-weighted processing cost, applied to every transaction	ST3 ERL scores for 85,187 transactions; dominant return reason from ST2 used to match each transaction to an intervention
Data used	85,187 completed transactions from the TheLook synthetic dataset; purchase-time features only	795,682 Amazon apparel reviews matched to TheLook product types using bigram matching; 275,499 labelled rows produced	ST1 and ST2 outputs joined on product type; no new data collected; 4 validation checks passed	ST3 ERL scores for 85,187 transactions; dominant return reason from ST2 used to match each transaction to an intervention
Main finding	Neither model predicted returns reliably on this dataset. A scenario score based on category averages was used instead.	Three of five reason categories (size_fit, defect, expectation_mismatch) were classified reliably. Overall macro F1 = 0.737.	ERL ranking shifted 95% of transactions more than 1,000 places compared with return-probability ranking. Gini improved by 51%.	Applying interventions to all matching transactions produced a net loss of £125,110. Targeting only high-risk profitable cases produced +£3.10.
Key limitation	The dataset had near-identical return rates across all product categories, so there was no useful signal for the model to learn from.	Two categories (changed_mind, wrong_item) had almost no usable training examples in the review corpus and could not be learned or poorly learnt.	All processing cost figures come from published research, not from this retailer. Actual costs may differ significantly.	The probability reduction figures used for each intervention are estimates from industry reports, not measured from a real pilot.
Next step	Re-run on real transaction data with a clearly defined return window and meaningful variation in return rates across categories.	Collect dedicated labelled examples for changed_mind and wrong_item rather than relying on the star-rating proxy approach.	Replace literature-based cost figures with retailer-specific values once real return processing data is available.	Pilot the top-performing intervention on a small sample and measure actual return rate changes before committing to full rollout.

Table 0: ERL project framework – Guidance for reader

v. List of Figures

Figure 1: ST1 pipeline output fields - primary dataset run	<i>page. 13</i>
Figure 2: Observed return rate by product category in eligible transactions	<i>page. 14</i>
Figure 3: Decile alignment of dataset-supplied p_return against return_label on the test partition	<i>page. 15</i>
Figure 4: Calibration of the selected model on the final test partition	<i>page. 15</i>
Figure 5: Sample output row handed to ST3 (one column per field) shows a sample row from the handoff file as passed to ST3.	<i>page. 17</i>
Figure 6: Average ERL by product category (left), ERL distribution boxplots (centre), and sensitivity analysis (right).	<i>page. 25</i>
Figure 7: Loss capture curve comparing ERL ranking, probability-only ranking, and random baseline	<i>page. 26</i>
Figure 8: ERL reason breakdown by product type (left: stacked bar; right: percentage heatmap).	<i>page. 28</i>
Figure 9: Sensitivity analysis — Net benefits of three targeting strategies across effectiveness scenarios.	<i>page. 34</i>

vi. List of Tables

Table 0: ERL project framework	<i>page. Vi</i>
Table 1: List of Abbreviations	<i>page. vii</i>
Table 2: Category weights used in the scenario return score.	<i>page. 11</i>
Table 3: Stage 1 output fields and their interpretation	<i>page. 12</i>
Table 4: Validation-set model comparison.	<i>page. 13</i>
Table 5: Final test assessment of selected model and diagnostic benchmarks	<i>page. 14</i>
Table 6: ST1 limitations, their impact, and required improvements.	<i>page. 16</i>
Table 7: ST1 inputs and outputs summarises the inputs, processing steps, and outputs of Stage 1.	<i>page. 17</i>
Table 8: Return reason taxonomy with definitions and representative lexical signals	<i>page. 18</i>
Table 9: Dataset roles across the ST2 pipeline.	<i>page. 18</i>
Table 10: Label generation summary from the full Amazon Apparel corpus.	<i>page. 20</i>
Table 11: Classification report on the 54,169-row held-out test set.	<i>page. 20</i>
Table 12: P(reason product_type) matrix, selected product types	<i>page. 21</i>
Table 13: ST2 inputs and outputs.	<i>Page. 22</i>
Table 14: Sample P(reason product_type) row handed to ST3 (Jeans)	<i>page. 22</i>
Table 15: Return cost matrix	<i>page. 23</i>
Table 16: ST3 headline results.	<i>page. 24/25</i>
Table 17: Financial exposure captured under different ranking strategies.	<i>page. 25</i>
Table 18: Gini coefficient comparison.	<i>page. 26</i>
Table 19: Risk tier summary across 85,187 transactions.	<i>page. 27</i>
Table 20: Limitations of Subtopic 3.	<i>page. 28</i>
Table 21: ST3 inputs and outputs.	<i>page. 28/29</i>
Table 22: Sample output row handed to ST4 (high-ERL transaction).	<i>page. 29</i>
Table 23: Approach A — Broad rollout versus the high-risk profitable filter	<i>page. 30/31</i>
Table 24: Approach B — Category-level net benefit by product group.	<i>page. 31</i>
Table 25: Summary of ST4 limitations, interpretive impact, and recommended improvements	<i>page. 32</i>
Table 26: Approach A - active scenario breakdown under broad rollout.	<i>page. 32</i>
Table 27: Approach B - product-type action plan, base case.	<i>page. 34</i>
Table 28: Sensitivity of Approach A	<i>page. 34</i>
Table 29: Sensitivity of Approach B	<i>page. 35</i>
Table 30: ST4 limitations, interpretive impact, and recommended improvements	<i>page. 36</i>

vii. List of Abbreviations

Abbreviation	Full Term
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BCS	British Computer Society
ERL	Expected Return Loss
F1	F1 Score (harmonic mean of precision and recall)
GDPR	General Data Protection Regulation
KL Divergence	Kullback–Leibler Divergence
ML	Machine Learning
NLP	Natural Language Processing
NIST	National Institute of Standards and Technology
NRF	National Retail Federation
SD	Standard Deviation
ST1	Sub-Topic 1: Return Probability Modelling
ST2	Sub-Topic 2: Return Reason Classification
ST3	Sub-Topic 3: ERL Construction
ST4	Sub-Topic 4: Intervention Evaluation
UK	United Kingdom

Table 1: List of Abbreviation

1. Introduction

Online fashion retail has the highest return rates of any consumer category. NRF (2024) reports that apparel returns exceed 24% of online sales in the United States. Roichman et al. (2024) document similar patterns in European markets, noting that returned apparel accounts for a disproportionate share of reverse logistics costs and greenhouse gas emissions. The financial impact goes beyond lost revenue. Each returned transaction triggers return shipping, inspection, restocking, and resale depreciation costs. These can exceed the original item margin on lower-value purchases. Despite this, the operational research literature has not developed a framework that connects return likelihood, return cause, and financial consequence at the level of the individual transaction.

The research literature addresses the returns problem in three separate streams. The first stream focuses on predicting whether a return will occur, using supervised machine learning on purchase-time features such as product category, customer history, and order value (Petersen and Kumar, 2009; Karl, 2025). The second stream uses natural language processing to identify why returns happen, drawing on customer review text and satisfaction signals (Ni et al., 2019; Malik and Bilal, 2024). The third stream estimates the operational costs of processing returns, including shipping, inspection, restocking, and markdown loss (Mishra and Dutta, 2024; Roichman et al., 2024). Karl (2025), reviewing 25 publications, finds that existing return forecasting research consistently emphasises volume prediction rather than transaction-level financial consequences. Reverse logistics optimisation models address some of these costs at an aggregate level (Mishra and Dutta, 2024), but none links return likelihood, return reason, and per-transaction cost estimation within a single scoring framework. No published framework brings all three streams together at the transaction level.

This project aims to develop, implement, and evaluate a transaction-level financial risk framework: the Expected Return Loss (ERL), that integrates return probability, return reason classification, and reason-specific processing cost into a single score, enabling retailers to prioritise return-prevention interventions by financial exposure rather than return likelihood alone. The formula $ERL_i = P(\text{return}_i) \times \sum [P(r|i) \times C(r)]$ expresses this as the product of three components: return probability from ST1, reason-conditional probability from ST2, and reason-specific cost from a literature-grounded cost matrix in ST3. Sub-Topic 4 then uses the ERL score to evaluate whether targeted interventions are financially justified. Each sub-topic is covered end-to-end in this report, with a handover paragraph connecting each

section to the next and a combined conclusion integrating findings across all four components.

2. LITERATURE REVIEW

Research on e-commerce returns has evolved in three largely independent streams: predicting returns, identifying the reasons for them, and estimating their financial impact. While each stream provides useful observations, current research rarely links the probability of a return, the reason for a return, and the cost of a return within a single framework for decision-making. A review of 25 published studies by Karl (2025) confirms that return forecasting research consistently prioritises volume prediction over financial consequence. Notably, no published framework has been found that integrates all three streams into a unified financial risk score at the transaction level. The literature review below highlights the contribution and limitations of each stream and motivates the development of the Expected Return Loss (ERL) framework as the first such integration.

2.1. Sub-topic 1: Return Probability Modelling

Return prediction is the most mature area of e-commerce returns research. Studies have demonstrated that return behaviour can be modelled by using purchase-time information such as product category, customer characteristics and order attributes (Petersen and Kumar, 2009; Nestler et al., 2021). However, Karl's (2025) review of 25 studies reveals a major limitation that performance often depends heavily on the dataset used, with limited evidence that models generalise across retail contexts, indicating that choosing a suitable dataset is as important as selecting the right algorithm. Because returns are typically a minority class, discrimination metrics such as AUC-ROC and Average Precision are preferred over accuracy (Hosmer et al., 2013; Saito and Rehmsmeier, 2015). AUC-ROC was selected as the primary discrimination metric because it measures a model's ability to rank returned transactions above non-returned transactions independently of any specific threshold. Average Precision was used as a complementary metric because it provides a more informative assessment of performance on the minority return class, where accuracy can be misleading due to class imbalance. Furthermore, calibration and discrimination should be assessed jointly to ensure probability estimates are both accurate and actionable (Niculescu-Mizil and Caruana, 2005; Zadrozny and Elkan, 2002). Hosmer et al. (2013, p.177) classify AUC-ROC values of 0.80 to 0.90 as excellent discrimination, establishing 0.80 as the threshold above which a classifier can be considered sufficiently reliable for downstream use. Where no classifier meets this threshold, a scenario-based score anchored to category-level return rates provides a documented fallback that preserves pipeline integrity without overstating predictive validity.

2.2. Sub-topic 2: Return Reason Classification

A major challenge in **return reason classification** is the lack of labelled return datasets. To address this, researchers increasingly rely on weak supervision approaches that use indirect signals to generate training labels. Ni et al. (2019) show that customer ratings are an effective measure of dissatisfaction, which allows large scale review corpora to be weakly labelled without any manual annotation. Recent findings show that transformer architectures consistently outperform traditional NLP methods on multi-class review classification tasks (Malik & Bilal, 2024). Natural Language Processing (NLP) helps in transforming the unstructured text of reviews into structured information for computation. One such model is DistilBERT that provides a good balance between performance and computational efficiency (Sanh et al., 2019). Being a transformer-based language model, it considers the context of words in customer reviews and enables us to classify return reasons more accurately than keyword-based approaches. Weak supervision frameworks such as Snorkel further improve scalability by combining multiple noisy labelling rules into probabilistic labels (Ratner et al., 2017). Where labels carry uncertainty across multiple classes, Hinton et al. (2015) demonstrate that training against soft probability distributions using KL Divergence loss preserves that uncertainty, whereas cross-entropy loss forces the model to treat ambiguous labels as confident single-class assignments and discards the information encoded in disagreement between labelling functions. Because return reasons differ substantially in their financial consequences, accurate and well-calibrated classification provides the critical link between return prediction and cost estimation.

2.3. Sub-topic 3: ERL Calculation

Existing return prediction research treats all returns as economically equivalent events. However, identical return probabilities can produce very different financial outcomes when return processing costs differ across categories. Cost-sensitive learning literature argues that prediction systems should account for the consequences of errors rather than accuracy alone (Elkan, 2001). In parallel, operations research identifies shipping, inspection, restocking and markdown losses as major components of return processing costs (Mishra and Dutta, 2024). Together, these studies suggest that financial exposure depends on both the likelihood of a return and the cost associated with its underlying cause.

Mishra and Dutta (2024) identify return shipping, inspection, restocking, and markdown loss as the four primary cost components of return processing. They note that reason category determines both the processing pathway and the cost level. A size complaint requires

restocking. A defect complaint requires inspection, possible repair, and often disposal or heavy markdown. An expectation mismatch may result in a full markdown if the item was accurately described but the customer found the images misleading. Reason-blind return modelling loses the cost differentiation that makes intervention decisions possible.

The Basel Committee on Banking Supervision (2004) defines expected financial loss as $EL = PD \times LGD \times EAD$, where PD is probability of default, LGD is loss given default, and EAD is exposure at default. This decomposition separates the likelihood of an adverse event from its financial magnitude. The formula is well-established in credit risk modelling. ST3 (Sub-Topic 3) adapts this structure to the retail returns domain, replacing PD with return probability from ST1 (Sub-Topic 1), and computing $LGD \times EAD$ as the sum of reason-weighted processing costs from ST2 (Sub-Topic 2) and the cost matrix. Unlike credit risk, where exposure at default is a fixed contractual value, retail return costs are reason-conditional and vary with product value: a defect return on a £900 item carries fundamentally different financial exposure than a size complaint on a £35 item, requiring a dynamic cost structure that the original Basel formulation does not provide. The cost matrix assigns a financial cost to each return reason category, allowing the expected cost of a return to be estimated based on the probability of different return outcomes.

2.4. Sub-topic 4: Intervention Evaluation

While prediction models identify potential returns, relatively little research considers whether **interventions** are financially worthwhile. Decision analysis literature argues that actions should be evaluated according to expected net benefit rather than prediction accuracy alone (Keeney and Raiffa, 1976). Industry evidence suggests that low-cost informational interventions, such as improved size guides and product descriptions, can reduce return rates more cost-effectively than operational interventions such as specialist callbacks or reship arrangements (IMRG, 2022; Narvar, 2019; Barclaycard, 2016). Building on Keeney and Raiffa's (1976) expected utility principle, this project operationalises intervention value as: $\text{net benefit} = (ERL \times \text{probability_reduction}) - \text{intervention_cost}$, where a positive result justifies deployment and a negative result does not. This formalisation extends the principle to the retail returns context by replacing abstract utility with a directly measurable financial quantity derived from the ERL score, enabling intervention decisions to be ranked and compared across product types and targeting strategies.

2.5. Project's Purpose

The three streams reviewed above have developed in isolation because each treats the return event differently: ST1 literature treats it as a prediction problem, ST2 literature treats it as a classification problem, and ST3 literature treats it as a cost accounting problem. Consequently, existing research provides no mechanism for prioritising transactions according to expected financial exposure. The ERL framework - our project - addresses this gap by integrating return probability, return reason classification, and reason-specific processing costs into a single transaction-level financial risk measure. The Basel adaptation in ST3 is not a cosmetic analogy: it represents a structural shift from fixed-exposure credit risk to reason-conditional, value-dependent retail risk, which required original methodological development.

2.6. Ethical and Fairness Considerations

A cross-cutting concern across all three research streams is the potential for algorithmically derived scores to produce uneven outcomes across demographic groups. If return rates, review text patterns, or purchasing behaviour differ systematically across customer segments, model outputs may vary in ways that aggregate metrics do not reveal (Tabassi, 2023). This risk is present in return probability models. It is also present in reason classifiers. And it is present in cost-weighted scoring systems. The relevant AI literature emphasises that predictive systems used in business decision-making should be screened for potential inequitable impacts prior to their implementation (BCS, 2022). This project uses exclusively synthetic and anonymized data and cannot directly assess such inequalities. However, the framework was developed with transparency and human oversight as explicit design principles, in line with the NIST framework for AI risk management (Tabassi, 2023).

3. Legal, Social, Ethical and Professional Issues

This project uses two publicly available datasets. TheLook is a synthetic e-commerce dataset generated by Google for research purposes. It contains no real individuals and involves no personal data. The Amazon Customer Reviews corpus is publicly available under a Creative Commons licence, was anonymised at the point of collection, and is used here only for aggregate text classification. No attempt is made to identify individuals from either source. Both datasets are used in compliance with their stated terms.

A production deployment on real customer data would require substantially more governance. Under UK GDPR, return probability scores derived from customer behaviour constitute inferred personal data. Processing them would require a lawful basis, data minimisation, and a data protection impact assessment before use. Transaction-level risk scoring at individual customer level would also engage Article 22 provisions on automated decision-making. If ERL scores were used to make decisions that significantly affect customers, such as restricting return rights or offering differential pricing, explicit consent or legitimate interest assessment would be required. ERL scores are financial prioritisation tools for internal operational use. They are not assessments of individual customer character or intent. Misuse in customer-facing decisions would be both ethically inappropriate and legally risky.

A secondary risk concerns algorithmic fairness. If return rates, review text patterns, or product purchasing behaviour differ systematically across demographic groups, model performance may vary across those groups in ways not visible in aggregate metrics. This project does not have access to demographic data and cannot test for such disparities. Any production deployment should evaluate model performance across customer segments before going live, and report disparities transparently. Categories with structurally different return drivers may require separate models rather than a single shared framework. The near-uniform return rates across categories observed in *Figure 2*(see *page 10*) illustrate this risk concretely: aggregate metrics can appear stable while underlying demographic variation remains invisible, reinforcing the case for segment-level evaluation before any production deployment.

The project follows documented-limitation reporting consistent with the NIST AI Risk Management Framework (Tabassi, 2023) and the BCS Code of Conduct (BCS, 2022). All cost assumptions are stated explicitly and tested through sensitivity analysis. The ST1 probability fallback is clearly labelled as unvalidated throughout the report. The ST2 residual categories are flagged in both the results and discussion sections. The framework is

designed to support human decision-making by providing financially grounded prioritisation. It is not designed to replace human judgement in operational or customer-facing contexts.

4. Methods: Framework Overview and Pipeline Design

This project addresses the research gap through a four-stage pipeline in which each stage produces a structured output that becomes a direct input to the next stage. The central research question cannot be answered by any single stage alone: only by integrating return probability, return reason, and reason-specific cost into a single score in order that the framework can produce financially actionable rankings. Therefore, the **core formula** driving the pipeline is: $ERL_i = P(\text{return}_i) \times \sum [P(\text{reason}_r | \text{product type}_i) \times C(\text{reason}_r)]$.

A sequential pipeline design was chosen over an end-to-end model for three reasons: interpretability at each stage allows individual component validation; the handover structure enables independent improvement of any single component without retraining the whole system; and the modular design allows ST1 and ST2 to be replaced with validated real-data versions without changing the ST3 or ST4 logic.

ST1 produces $P(\text{return}_i)$. ST2 produces the $P(\text{reason} | \text{product type})$ distribution. ST3 constructs the cost matrix $C(\text{reason})$ from published literature and assembles all three components into a transaction-level ERL score. Lastly, ST4 uses those scores to evaluate whether specific interventions are worth deploying.

All four sub-topics draw from a common base of 85,187 completed transactions from the ‘**TheLook**’ **synthetic e-commerce dataset**, that is provided by Google for research purposes, containing realistic but entirely fictional retail transaction records with no real personal data. ST2 additionally draws on 795,682 Amazon apparel reviews filtered to one- and two-star ratings. Those are matched to TheLook product types by using bigram matching and solely for text classification training. ST3 and ST4 introduce no new datasets and operate entirely on the structured outputs passed down from ST1 and ST2.

ST1 trains a Logistic Regression and Random Forest classifiers on TheLook transaction data using only purchase-time features to prevent data leakage. A pre-defined validation threshold of $AUC-ROC \geq 0.80$ determines whether the classifier output or a scenario-based fallback score is passed to ST3. On the primary dataset, neither classifier met this threshold and the scenario fallback was used.

ST2 operates in parallel with ST1, answering a different question from ST1: not whether a return will happen, but why. It applies Snorkel weak supervision to generate training labels across five return reason categories and fine-tunes a DistilBERT classifier on those labels. The output is a $P(\text{reason} | \text{product type})$ matrix covering 24 product types, which is the second input to ST3.

ST3 integrates those outputs of ST1 and ST2. It joins the ST1 return probabilities and ST2 reason distributions with a literature-grounded cost matrix. Then applies the ERL formula to all 85,187 transactions and ranks them by both ERL and return probability independently and assigns each to a percentile-based risk tier. The resulting 22-column output file is the sole input to ST4.

ST4 evaluates whether targeted return-prevention interventions are financially justified, operating at two levels. A transaction-level model matches each transaction to one of twelve intervention scenarios and computes net benefit as ERL saving minus intervention cost. Furthermore, a category-level model pools ERL exposure by product type and evaluates strategic pre-transaction interventions at scale. The two-level design was necessary because per-transaction ERL averaged £2.20, making individual-level intervention rarely cost-effective while the same logic became viable when applied across product groups.

3. Sub-Topic 1: Return Probability Modelling

3.1. Methods and Data

ST1 treats return prediction as a binary classification task. The target variable is `return_label`, which records whether an eligible completed purchase resulted in a return. Predictors are restricted to information available at purchase time: `product_category`, `product_type`, `brand`, `order_value_pound`, `num_of_item`, and temporal features derived from `order_date` (month, `day_of_week`, `is_weekend`). Post-purchase fields are excluded to prevent data leakage. The dataset-supplied `p_return` field is excluded from model training and evaluated separately as a diagnostic reference.

Records are ordered chronologically and split 60% training, 20% validation, and 20% testing, yielding 51,112 training rows, 17,037 validation rows, and 17,038 test rows. Chronological ordering prevents future information leaking into training.

Two classifiers are evaluated: Logistic Regression as a transparent baseline and Random Forest to test whether non-linear feature interactions improve discrimination. The two-classifier design creates a principled diagnostic test: if both a linear and a non-linear model produce near-chance results independently, that provides stronger evidence of absent signal than a single model could. Both use balanced class weights to address class imbalance. Predicted probabilities are adjusted using Platt scaling (Platt, 2000) to improve calibration.

Product Category	Scenario Weight
Occasionwear	1.30
Denim & Bottoms	1.15
Swimwear & Active	1.10
Knitwear & Tops	1.00
Premium Outerwear	0.90
Basics & Hosiery	0.85
Intimates & Nightwear	0.60
Accessories	0.55

Table 2: Category weights used in the scenario return score.

Validation AUC-ROC of at least 0.80 is the pre-defined requirement for handoff to ST3. This threshold is set above the 0.70 acceptable floor from Hosmer et al. (2013) because a score entering a shared downstream pipeline requires stronger evidence than a general

acceptability standard. If neither model meets this threshold, `p_return_final` is populated using a `scenario_return_score`. The training-period return rate multiplied by an analyst-defined category weight (*Table 2*), which maintains the numeric format required downstream without claiming predictive validity. The dataset contains 85,187 eligible completed transactions with an overall return rate of 11.8%.

3.2. Implementation

Preprocessing is applied within the modelling pipeline to prevent data leakage. Missing categorical values are assigned a placeholder before one-hot encoding. Missing numeric values are imputed using the training-period median. Numeric features are standardised. All transformations are fitted only on the training partition, preventing validation or test data from influencing preprocessing. Both classifiers are fitted with Platt scaling.

The selected classifier is assessed once on the held-out test partition after model selection is complete, separating model selection from final performance reporting.

Field	Type	Interpretation
<code>p_return_final</code>	Float [0,1]	Handoff field. Equals <code>scenario_return_score</code> in this run. Not a validated prediction.
<code>selected_ml_probability</code>	Float [0,1]	Calibrated model output retained for evaluation and audit only.
<code>scenario_return_score</code>	Float [0,1]	Assumption-based score anchored to training-period return rate.

Table 3: Stage 1 output fields and their interpretation

The dataset-supplied `p_return` field is assessed separately using the same metrics and a decile diagnostic, which checks whether transactions in the highest-scoring decile have materially higher observed return rates than those in the lowest. Three output fields are assembled for handoff (*Table 3*): `p_return_final` carries the value used downstream; `selected_ml_probability` retains the calibrated model output for audit; and `handoff_input_type` records which path was taken.

	transaction_id	product_category	order_value_pound	return_label	selected_ml_probability	scenario_return_score	p_return_final
0	2	Denim & Bottoms	125.00	0	0.1153	0.1356	0.1356
1	5	Denim & Bottoms	59.99	0	0.1168	0.1356	0.1356
2	6	Intimates & Nightwear	108.75	0	0.1153	0.0707	0.0707
3	7	Swimwear & Active	109.99	0	0.1152	0.1297	0.1297
4	8	Denim & Bottoms	59.50	0	0.1153	0.1356	0.1356
5	9	Swimwear & Active	63.97	0	0.1158	0.1297	0.1297
6	10	Knitwear & Tops	148.00	0	0.1195	0.1179	0.1179
7	15	Knitwear & Tops	44.99	0	0.1159	0.1179	0.1179
8	14	Accessories	99.83	0	0.1159	0.0648	0.0648
9	16	Premium Outerwear	84.21	0	0.1165	0.1061	0.1061

Figure 1: ST1 pipeline output fields - primary dataset run

Figure 1 gives exported values. Because the ML probability fails validation, *p_return_final* carries *scenario_return_score* rather than *selected_ml_probability*.

To confirm the pipeline works correctly when discriminative signal exists, the notebook was re-run on an alternative synthetic dataset. This dataset is structurally identical to the primary one: same schema, same number of records, and same exclusion logic but was adjusted solely to introduce meaningful variation in return rates across product categories, simulating a deployment context where discriminative signal exists. Return rates range from 1.9% to 59.9% across categories (SD = 0.198, versus 0.004 on the primary dataset). Random Forest achieved validation AUC-ROC of 0.807 on this dataset, satisfying the threshold. The test partition confirmed performance at 0.809 (Average Precision 0.553, Brier Score 0.154). The *handoff_input_type* field read *calibrated_ml_probability*, confirming the correct branch was taken. Full results are in Appendix A. This demonstration establishes that the near-chance performance on the primary dataset is a consequence of that dataset's structure, not a fault in the pipeline code.

3.3. Results

Model	AUC-ROC	Avg Precision	Brier	Meets >= 0.80?
Logistic Regression	0.5063	0.1212	0.1057	No
Random Forest	0.4997	0.1212	0.1057	No

Table 4: Validation-set model comparison.

Neither classifier reached the pre-defined threshold, as shown in *Table 4*. Logistic Regression achieved validation AUC-ROC of 0.5063 and Random Forest achieved 0.4997, both at chance level. Logistic Regression was carried forward to the final test partition, where it achieved AUC-ROC of 0.4919, close to the base-rate benchmark of 0.5000. The

dataset-supplied p_return field achieved 0.5001. Full test results and diagnostic benchmarks are reported in *Table 5*.

Model / Benchmark	AUC-ROC	Avg Precision	Brier	Interpretation
Logistic Regression (test)	0.4919	0.1205	0.1059	Reported once on held-out partition
Training base-rate baseline	0.5000	0.1204	0.1059	Reference benchmark
Dataset p_return (diagnostic)	0.5001	0.1191	0.1267	Non-monotonic deciles; retained for audit only

Table 5: Final test assessment of selected model and diagnostic benchmarks.

On the held-out test partition, Logistic Regression achieved AUC-ROC of 0.4919, close to the base-rate benchmark of 0.5000 (*Table 5*). The dataset-supplied p_return field achieved 0.5001. Both are effectively at chance level.

return rate by category

product_category	return_rate	n_returns	n_total
Basics & Hosiery	0.1219	482	3953
Premium Outerwear	0.1217	1639	13468
Swimwear & Active	0.1202	1269	10560
Denim & Bottoms	0.1180	2649	22455
Knitwear & Tops	0.1159	2178	18785
Occasionwear	0.1156	441	3815
Accessories	0.1134	295	2601
Intimates & Nightwear	0.1127	1076	9550

Figure 2: Observed return rate by product category in eligible transactions

As shown in *Figure 2*, category-level return rates ranged from 11.3% (Accessories) to 12.2% (Basics and Hosiery), a range of less than one percentage point. With near-identical return rates across all categories, the categorical features that would normally provide most of the discriminative power in a return model carry almost no information on this dataset. Accordingly, p_return_final is populated by the scenario score for all 85,187 exported records.

	decile	mean_p_return	observed_return_rate	n
0	(0.00039999999999999996, 0.0598]	0.038587	0.118406	1706
1	(0.0598, 0.0924]	0.076441	0.119718	1704
2	(0.0924, 0.122]	0.107454	0.112155	1703
3	(0.122, 0.15]	0.135772	0.122797	1702
4	(0.15, 0.18]	0.165083	0.126316	1710
5	(0.18, 0.212]	0.196036	0.115814	1701
6	(0.212, 0.251]	0.231237	0.129336	1701
7	(0.251, 0.296]	0.272390	0.132120	1703
8	(0.296, 0.367]	0.327558	0.114302	1706
9	(0.367, 0.758]	0.445141	0.112808	1702

Figure 3: Decile alignment of dataset-supplied p_return against $return_label$ on the test partition

The decile diagnostic confirmed the lack of discriminative signal. As shown in *Figure 3*, the mean predicted p_return rises from 0.039 in the lowest decile to 0.445 in the highest. Observed return rates remain flat at approximately 11 to 13% across all deciles. Higher model scores do not correspond to higher observed return rates. This confirms the models have not learned to distinguish returning from non-returning transactions.

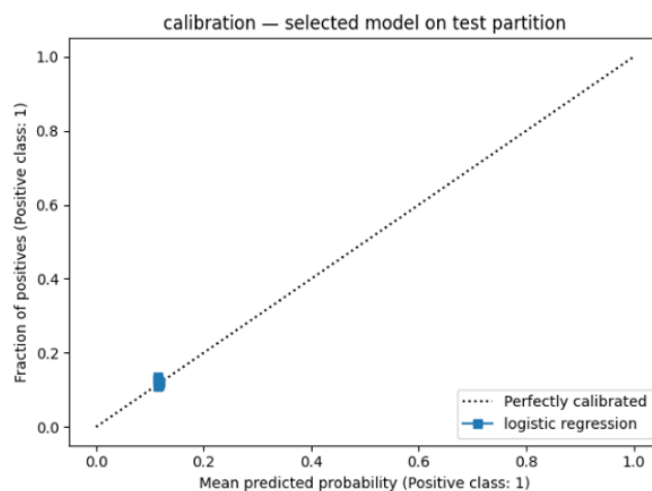


Figure 4: Calibration of the selected model on the final test partition

Figure 4 shows selected-model probabilities concentrated near the overall return rate rather than across a meaningful risk range. This does not demonstrate useful targeting: a model can reflect the base rate while failing to discriminate returned from non-returned purchases.

3.4. Discussion and Limitations

The data do not support validated ML return prediction, confirmed by three independent signals: category return rates are near-flat (SD = 0.004); the dataset-supplied `p_return` field fails independently at AUC-ROC 0.5001; and both a linear and a non-linear classifier produce near-identical chance-level results. Their convergence makes it unlikely that a different modelling choice, including gradient boosting, would have succeeded, as no classifier can extract signal the data structurally lack (Karl, 2025).

The scenario fallback is a pipeline placeholder, not a validated probability. Category weights conflict with observed rates in several cases, for example, Basics and Hosiery carries a weight of 0.85 despite the highest observed return rate at 12.2%, confirming that the weights reflect assumed rather than measured risk. ERL outputs in this run should therefore be read as scenario-weighted illustrations of the pipeline structure rather than transaction-level financial risk assessments. The full set of limitations, their interpretive impact, and required improvements are summarised in *Table 6*. Future work should re-evaluate the pipeline on real historical data with meaningful cross-category return rate variation; the deployment validation run in Appendix A confirms the pipeline is architecturally sound for this purpose.

Limitation	Impact on Interpretation	Required Improvement
Synthetic dataset only	Relationships may not reflect real retail returns	Re-evaluate on representative historical transaction and return data
Return horizon undefined	Target is not return within a specified policy window	Define a return window (e.g. 30 days) before predictive use
Scenario weights not validated	<code>p_return_final</code> is not a validated probability in this run	Estimate or validate weights using suitable real data
Calibration not time-aware	Prototype is not fully chronological at every modelling step	Use time-aware calibration and repeat evaluation

Table 6: ST1 limitations, their impact, and required improvements.

ST1 exports a return-likelihood score per transaction, with inputs, processing steps, and outputs summarised in *Table 7* and a sample output row shown in *Figure 5*. In this run that score is the unvalidated scenario fallback. It occupies the `P(return)` slot in the ERL formula but must not be read as a validated probability. Sub-Topic 2 produces the reason distribution needed to complete the financial risk estimate.

Detail	
Input dataset	TheLook synthetic transactions: 85,187 rows after cancellations removed
Input features	product_category, product_type, brand, order_value_pound, num_of_item, month, day_of_week, is_weekend
Processing	Chronological 60/20/20 split; Logistic Regression + Random Forest with Platt scaling; AUC threshold check; scenario fallback if threshold not met
Key output field	p_return_final: return-likelihood score
Output file	output_p_return.csv: 85,187 rows, joined by transaction_id

Table 7: ST1 inputs and outputs | summarises the inputs, processing steps, and outputs of Stage 1.

transaction_id	order_id	customer_id	product_category	product_type	brand	order_value_pound	num_of_item	return_label	selected_ml_probability	scenario_return_score	p_return_final	handoff_input_type	p_return_data_set_reference
2	1	1	Denim & Bottoms	Jeans	Joe's Jeans	125.00	3	0	0.1153	0.1356	0.1356	scenario_score_unvalidated	0.2472
5	3	3	Denim & Bottoms	Pants	Dockers	59.99	1	0	0.1168	0.1356	0.1356	scenario_score_unvalidated	0.1642
6	4	3	Intimates & Nightwear	Sleep & Lounge	Tommy Bahama	108.75	1	0	0.1153	0.0707	0.0707	scenario_score_unvalidated	0.3175
7	5	3	Swimwear & Active	Active	SmartWool	109.99	1	0	0.1152	0.1297	0.1297	scenario_score_unvalidated	0.1088
8	6	3	Denim & Bottoms	Jeans	Marc Ecko Cut & Sew	59.50	2	0	0.1153	0.1356	0.1356	scenario_score_unvalidated	0.3638
9	6	3	Swimwear & Active	Swim	Volcom	63.97	2	0	0.1158	0.1297	0.1297	scenario_score_unvalidated	0.0726
10	7	4	Knitwear & Tops	Sweaters	Lilly Pulitzer	148.00	1	0	0.1195	0.1179	0.1179	scenario_score_unvalidated	0.3821
15	9	6	Knitwear & Tops	Fashion Hoodies & Sweatshirts	Quiksilver	44.99	2	0	0.1159	0.1179	0.1179	scenario_score_unvalidated	0.1171
14	9	6	Accessories	Accessories	Ray-Ban	99.83	2	0	0.1159	0.0648	0.0648	scenario_score_unvalidated	0.1274
16	10	7	Premium Outerwear	Suits & Sport Coats	Haggar	84.21	1	0	0.1165	0.1061	0.1061	scenario_score_unvalidated	0.2445
25	14	11	Swimwear & Active	Swim	Becca by Rebecca Virtue	132.00	1	1	0.1168	0.1297	0.1297	scenario_score_unvalidated	0.0898

Figure 5: Sample output row handed to ST3 (one column per field) | shows a sample row from the handoff file as passed to ST3.

4. Sub-Topic 2: Return Reason Classification

4.1. Methods and Data

ST2 produces a $P(\text{reason} \mid \text{product_type})$ matrix giving the probability distribution across five return reasons for each product type. This matrix is the second input to the ERL formula in ST3. Five reason categories were defined to cover the dominant drivers of fashion returns identified in the literature, each anchored by a written definition and representative lexical signals, as shown in *Table 8*.

Category	Definition	Typical Lexical Signal
size_fit	Item did not fit or was the wrong size	"too small", "runs large", "size chart misleading"
defect	Item was damaged, broken, or poor quality	"fell apart", "cheap quality", "stitching came undone"
expectation_mismatch	Item did not match description, photos, or dimensions	"not as described", "colour is off", "nothing like the picture"
wrong_item	Customer received a different product than ordered	"received the wrong item", "sent the wrong one"
changed_mind	Customer no longer wanted the item	"changed my mind", "ordered by mistake", "no longer need"

Table 8: Return reason taxonomy with definitions and representative lexical signals.

Bigram matching was used to assign Amazon reviews to TheLook product types, achieving 93.5% coverage of unique TheLook product names, compared with 93% for keyword matching and approximately 18% for trigram matching. The 6.5% of Amazon titles that could not be matched were excluded from the output matrix. The three datasets supporting the pipeline and their roles are described in *Table 9*.

Dataset	Size	Purpose
124k calibration corpus	124,979 rows	Tunes labelling functions. Never used for model training.
33k primary corpus	33,835 rows	TheLook products matched to Amazon reviews. Produces the $P(\text{reason} \mid \text{product_type})$ matrix for ST3.
Full Amazon Apparel TSV	795,682 rows (1-2 star filtered)	Snorkel produced 275,499 high-confidence training labels for DistilBERT.

Table 9: Dataset roles across the ST2 pipeline.

4.2. Implementation

The pipeline ran across six phases. **Phase 0** assigned Amazon reviews to TheLook product types using bigram matching, selected after keyword and trigram approaches were found to have lower coverage. **Phase 1** loaded and validated the 33k and 124k datasets, concatenated review headline and body into a single text column, and excluded reviews under 20 characters as uninformative.

Phase 2 developed 23 labelling functions in Python using the Snorkel framework, encoding domain knowledge through phrase-based matching, co-occurrence functions, negation detection, and star-rating gates. A co-fire function prevented a general return-indicating phrase from assigning a default category label. The label model was calibrated on the 124k corpus before application to the full Amazon TSV.

Phase 3 applied the frozen label model to the 795,682-row Amazon Apparel TSV filtered to 1–2 star reviews, producing 275,499 high-confidence labelled rows at confidence threshold 0.65 or above. A keyword-extraction fallback assigned hard labels for `wrong_item` and `changed_mind`, whose calibration corpus coverage fell below 0.03%. After deduplication, the training dataset contained 270,842 rows.

Phase 4 validated labels through manual annotation of a stratified sample of 245 reviews by two independent annotators, achieving inter-rater agreement of Kappa = 0.91 — classified as almost perfect by Landis and Koch (1977). Snorkel agreement with the human reference was 82.4%, exceeding the 70% stopping threshold required before model training.

Phase 5 fine-tuned DistilBERT on a stratified 20,000-row sample drawn due to CPU memory constraints, using batch size 16, learning rate 2×10^{-5} , maximum 4 epochs, and early stopping on validation loss. KL Divergence replaced cross-entropy as the loss function to preserve uncertainty in Snorkel's soft probability labels. **Phase 6** evaluated the saved checkpoint on the held-out test set of 54,169 reviews and applied it to all 33,835 primary dataset rows, grouping predictions by `product_type` to produce the $P(\text{reason} \mid \text{product_type})$ matrix exported to ST3. A diagnostic experiment augmenting the model with keyword-extracted hard labels for `wrong_item` and `changed_mind` achieved F1 of 0.87 and 0.92, but these scores are optimistic upper bounds since the test set was drawn from the same keyword-extracted pool; the Phase 5 model is used in production.

4.3. Results

The model was tested on 54,169 held-out reviews. The overall macro F1 of 0.737 exceeds the project target of 0.68, as shown in *Table 11*. Label generation counts by category are reported in *Table 10*. Three categories performed reliably and two did not.

Category	High-Confidence Labels (≥ 0.65)	% of Total Processed
defect	147,550	18.5%
size_fit	116,812	14.7%
expectation_mismatch	9,717	1.2%
wrong_item	1,347	0.2%
changed_mind	73	0.009%
Total	275,499	34.6%

Table 10: Label generation summary from the full Amazon Apparel corpus.

Category	Precision	Recall	F1	Support	Meets Target (≥ 0.68)
size_fit	0.965	0.964	0.965	23,106	Yes
defect	0.967	0.973	0.970	28,860	Yes
expectation_mismatch	0.902	0.861	0.881	1,921	Yes
wrong_item	0.939	0.809	0.869	267	Yes
changed_mind	0.0	0.0	0.0	15	No
Macro Average	0.755	0.721	0.737	54,169	Yes

Table 11: Classification report on the 54,169-row held-out test set.

size_fit and defect achieved F1 of 0.965 and 0.970 respectively, reflecting the most training examples and the clearest language patterns. expectation_mismatch achieved 0.881 despite being the most linguistically ambiguous category. wrong_item achieved 0.869 despite only 267 test examples, reflecting the unambiguous nature of wrong-item complaints. changed_mind scored 0.0, only 15 test examples were available, far too few to measure reliably. Table 10 explains why: changed_mind produced only 73 high-confidence labels from 795,682 reviews, compared with 147,550 for defect. Customers who change their mind rarely write detailed 1-star reviews, so scaling the corpus did not close the gap.

The trained model was applied to all 33,835 primary dataset rows to produce the output matrix shown in *Table 12*. `size_fit` dominates for tailored garments, Blazers and Jackets scored 0.630 and Plus scored 0.635. `defect` probability clustered between 0.30 and 0.43 across most categories. `Changed_mind` shows near-flat variation across all 24 product types, confirming it is a softmax residual rather than a learned prediction. `wrong_item` shows similarly low variation, but its F1 of 0.869 on the test set indicates a partially learned signal despite sparse training data; the flat matrix output reflects the limited number of `wrong_item` reviews in the corpus rather than a complete absence of learning.

Product Type	size_fit	defect	expect. mismatch	wrong_item	changed_mind
Blazers & Jackets	0.630	0.218	0.062	0.031	0.059
Plus	0.635	0.124	0.099	0.054	0.088
Jeans	0.437	0.305	0.109	0.054	0.096
Outerwear & Coats	0.452	0.310	0.101	0.049	0.087
Accessories	0.271	0.428	0.126	0.071	0.104
Sleep & Lounge	0.348	0.381	0.111	0.062	0.098
Tops & Tees	0.386	0.342	0.115	0.059	0.097

Table 12: $P(\text{reason} \mid \text{product_type})$ matrix, selected product types.

4.4. Discussion and Limitations

The three reliable classes: `size_fit`, `defect`, and `expectation_mismatch`, cover the dominant apparel return drivers and carry the most differentiated cost structures in ST3. A `size_fit` complaint costs £10.00 to process; a `defect` complaint costs £15.00 plus 20% of order value; an `expectation_mismatch` complaint costs £18.00 flat. Correct classification of these three categories is what gives the ERL formula its financial differentiation power.

The `changed_mind` and `wrong_item` failures are structural consequences of the distant supervision approach: the proxy labelling assumption holds strongly for quality and fit complaints but breaks down for behavioural and fulfilment complaints, where customers rarely write detailed low-star reviews. F1 scores are measured against Snorkel labels rather than human ground truth; with Snorkel agreement at 82.4%, approximately 1 in 5 labels may be incorrect. Training used 20,000 rows rather than the full 270,842 due to CPU constraints, which likely limits performance on `expectation_mismatch` and `wrong_item`. Future work

should collect dedicated labelled data for the two residual categories. Full limitations are summarised in *Table 13*.

*ST2 exports the $P(\text{reason} \mid \text{product_type})$ matrix, with inputs, processing steps, and outputs summarised in [Table 13](#) and a sample output row for Jeans shown in [Table 14](#). These distributions are combined with the per-transaction return probabilities from *ST1* and the reason-specific cost structure in *ST3* to compute the Expected Return Loss for every transaction.*

Detail	
Input (training)	Full Amazon Apparel TSV filtered to 1-2 star: 795,682 rows; 275,499 high-confidence Snorkel labels at confidence ≥ 0.65
Input (output)	33k primary corpus: 33,835 TheLook-matched Amazon reviews; bigram-matched to 24 product types
Processing	Snorkel label model (23 labelling functions); DistilBERT fine-tuned with KL Divergence loss; 20,000-row stratified training sample
Key output	$P(\text{reason} \mid \text{product_type})$ matrix: 24 rows x 5 reason probability columns
Output file	table10_p_reason_product_type.csv: 24 rows, joined to ST3 on product_type

Table 13: *ST2 inputs and outputs.*

product_type	size_fit	defect	expectation_mismatch	wrong_item	changed_mind
Jeans	0.437	0.305	0.109	0.054	0.096

Table 14: *Sample $P(\text{reason} \mid \text{product_type})$ row handed to ST3 (Jeans).*

5. Sub-Topic 3: ERL Construction

5.1. Methods and Data

The ERL formula is: $ERLi = P(\text{return}i) \times \sum_r [P(r|i) \times C(r)]$. $P(\text{return}i)$ comes from ST1, $P(r|i)$ comes from ST2, and $C(r)$ is the reason-specific cost from the cost matrix in *Table 15*. Most costs are flat rates reflecting typical handling activities independent of item value. Defect cost is value-dependent because resale depreciation and inspection cost scale with item price, computed as $\text{£}15.00 + 0.20 \times \text{order_value_pound}$, ranging from $\text{£}22.20$ to $\text{£}214.80$ across the dataset.

Return Reason	Cost Formula	Type	Source
Size/fit	£10.00	Flat	Mishra and Dutta (2024)
Defect	£15.00 + 20% x order value	Value-dependent	Roichman et al. (2024)
Expectation mismatch	£18.00	Flat	NRF (2024)
Wrong item	£22.00	Flat	Roichman et al. (2024)
Changed mind	£8.00	Flat	Mishra and Dutta (2024)
Return shipping	£3.85 (weighted avg)	Empirical	Royal Mail (2024)

Table 15: Return cost matrix. | presenting the cost assigned to each return reason category, sourced from published literature except for return shipping, which was estimated empirically from the actual order value distribution of the dataset.

ST3 introduces no independent dataset. It integrates ST1 outputs (85,187 transactions with return probabilities and order values) and ST2 outputs (reason distributions for 24 product types) via a left join on `product_type`. Two product types absent from ST2 output: Clothing Sets (159 transactions) and Suits and Sport Coats (3,822 transactions), were imputed using the mean distribution across all 24 covered types, affecting 4.7% of the dataset. Four validation checks confirmed data integrity before any calculation: no missing values in required columns; all `p_return` values within $[0,1]$; all order values positive; and all reason probability rows summing to 1.0 within tolerance. All four checks passed.

5.2. Implementation

Stages 1 and 2 loaded and merged the ST1 and ST2 outputs on product_type and confirmed all four validation checks passed. Stage 3 estimated return shipping costs empirically using Royal Mail Tracked Returns pricing tiers: £2.99 (under £50), £3.99 (£50–£150), £4.99 (£150–£300), and £5.99 (over £300), yielding a weighted average of £3.85 across the 10,029 returns in the dataset. Stage 4 applied the cost matrix. Stage 5 computed the weighted cost $\text{SUM}[P(r|i) \times C(r)]$ across all five reasons per transaction, with a mean of £18.98 (SD £5.09), then multiplied by $P(\text{return})$ to produce ERL_base. Optimistic and pessimistic variants were produced by scaling costs by 0.75 and 1.25 respectively. Stage 6 ranked all transactions by both p_return_final and ERL_base independently, computed the Gini coefficient for each ranking, and generated the loss capture curve. Stage 7 classified transactions into five percentile-based risk tiers. Stage 8 exported the 85,187-row, 22-column output file in both CSV and Excel format, as summarised in Table 21.

5.3. Results

Base ERL scores ranged from £1.08 to £11.81, with a mean of £2.20 (SD £0.73) and a median of £2.15. The distribution is right-skewed, confirming a high-risk tail of high-value transactions with elevated defect exposure. Including empirically estimated shipping costs raised total expected exposure to £225,310.98. The predicted exposure captured 98.1% of actual realised losses of £190,998.53. Headline results are summarised in *Table 16*, and *Figure 6* shows average ERL by product category, the ERL score distribution, and the sensitivity analysis under optimistic and pessimistic cost assumptions.

Metric	Value
Transactions analysed	85,187
Overall return rate	11.8%
Average weighted cost	£18.98
Average ERL (base)	£2.20
Total ERL exposure (base)	£187,357.98
Total ERL exposure (with shipping)	£225,310.98
Highest ERL	£11.81
Transactions shifting >1,000 ranks	95.0% (80,968 transactions)

Top 10% exposure captured (ERL)	16.6%
Top 10% exposure captured (probability)	12.5%

Table 16: ST3 headline results.

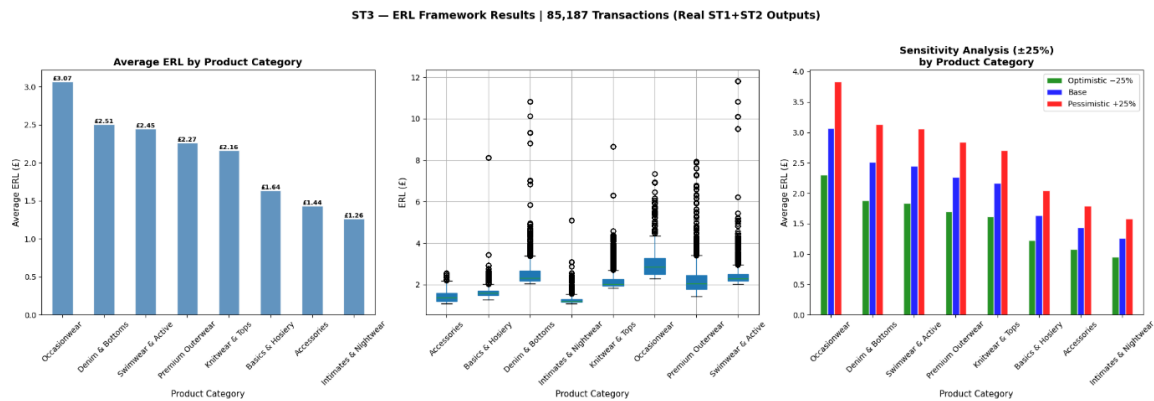


Figure 6: Average ERL by product category (left), ERL distribution boxplots (centre), and sensitivity analysis (right).

The ranking comparison identified 80,968 transactions (95.0%) where ERL and probability rankings diverged by more than 1,000 positions, as shown in *Table 17*. The most extreme case was a Sleep and Lounge item valued at £399.95 with $P(\text{return}) = 0.108$, which shifted 66,964 positions because its defect exposure at £95.00 makes it far more financially significant than its moderate return probability suggests. *Figure 7* shows the loss capture curve, confirming ERL ranking consistently captures more financial exposure than probability ranking across all targeting thresholds.

Top % Targeted	ERL Captures	Probability Captures	Random Captures	ERL Improvement
5%	9.49%	6.84%	5.01%	+2.65 pp
10%	16.60%	12.52%	10.00%	+4.08 pp
20%	29.00%	23.94%	20.03%	+5.06 pp
30%	40.10%	35.36%	30.02%	+4.74 pp

Table 17: Financial exposure captured under different ranking strategies.

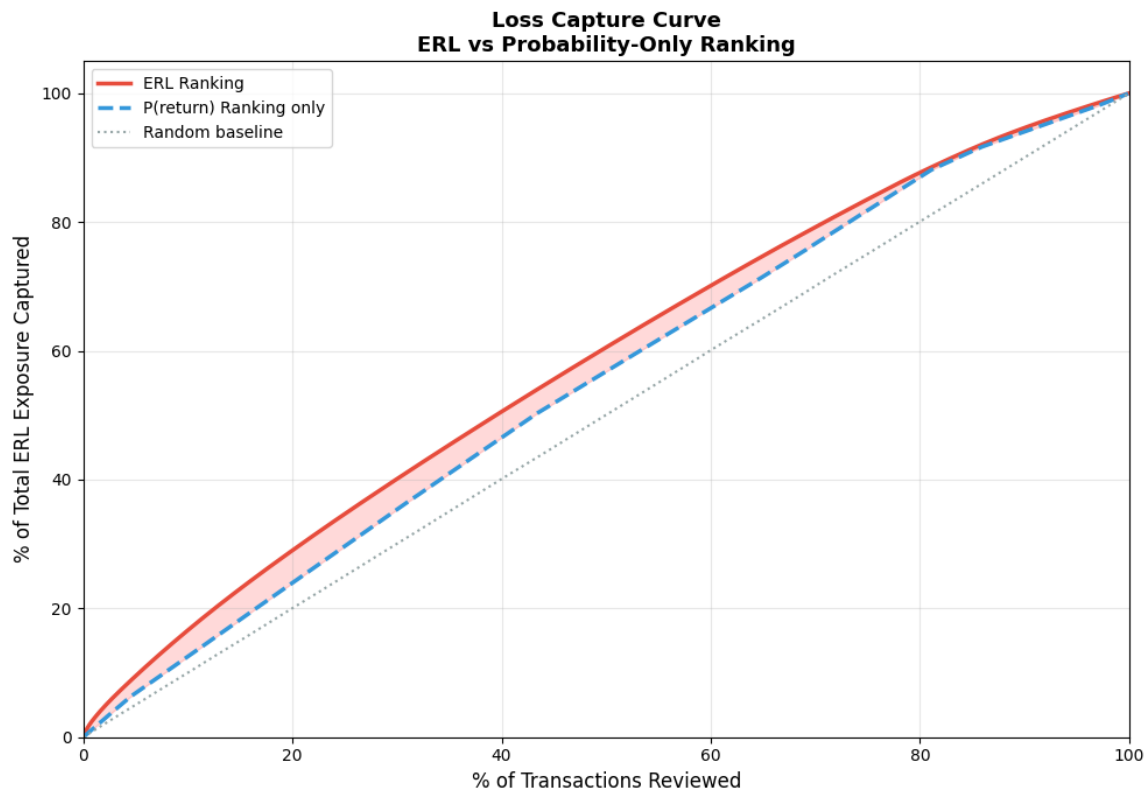


Figure 7: Loss capture curve comparing ERL ranking, probability-only ranking, and random baseline.

Metric	Value	Meaning
Gini (ERL)	0.1601	ERL ranking concentrates financial exposure more effectively
Gini (P(return))	0.1058	Probability-only ranking is less financially concentrated
Relative improvement	1.51x	ERL almost doubles concentration efficiency

Table 18: Gini coefficient comparison.

As shown in *Table 18*, the Gini coefficient rose from 0.1058 under probability-only ranking to 0.1601 under ERL ranking: a 51% improvement in concentration efficiency. The risk tier analysis in *Table 19* confirmed that return rates across tiers are near-uniform at 11.6% to 12.2%, confirming that ERL tiers reflect cost concentration rather than return likelihood. A high-ERL transaction does not return more often than a low-ERL transaction. It costs far more when it does.

Risk Tier	Transactions	Return Rate	Avg ERL	Total ERL	% of Total ERL
Low	42,675	11.6%	£1.74	£74,114.90	39.6%
Medium	21,528	11.7%	£2.28	£49,060.56	26.2%
High	12,589	12.2%	£2.66	£33,447.63	17.9%
Very High	4,250	12.2%	£3.14	£13,340.58	7.1%
Critical	4,145	11.8%	£4.20	£17,394.31	9.3%

Table 19: Risk tier summary across 85,187 transactions.

Defect-related losses were the dominant financial driver across most product types, contributing 50–67% of total ERL. Outerwear and Coats derived 65% of its exposure from defect risk; Blazers and Jackets and Plus derive over 60% from size_fit. *Figure 8* illustrates this reason-level breakdown by product type, and the full risk tier decomposition by reason is provided in Appendix D.

5.4. Discussion and Limitations

The 95% ranking divergence is the central result of ST3, consistent with Elkan (2001), who demonstrates that cost-blind models systematically misallocate attention. The loss capture results quantify the practical value: at the top 10% targeted, ERL captures 16.6% of total financial exposure versus 12.5% for probability ranking: a 4.1-percentage-point gap that at one million transactions per year represents an additional £41,000 of recoverable exposure before any intervention cost is considered. The reason-level decomposition, shown in *Figure 8*, makes category-level differentiation possible: different product types require different intervention strategies rather than a single uniform approach.

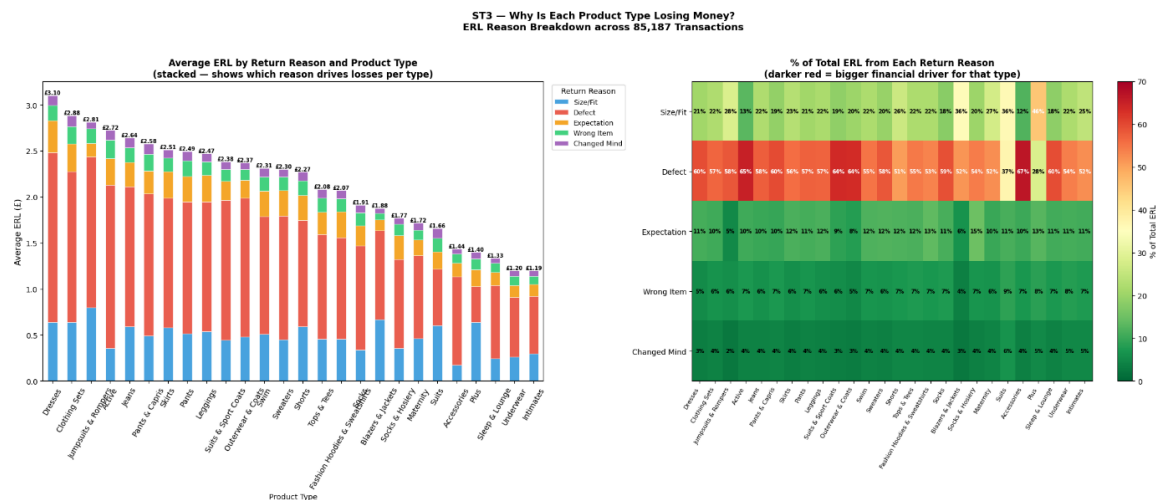


Figure 8: ERL reason breakdown by product type (left: stacked bar; right: percentage heatmap).

Limitation	Impact on Interpretation	Required Improvement
Literature-based cost estimates	Actual costs may differ significantly from modelled values	Replace with retailer-specific processing records
Narrow ST1 probability spread (0.065-0.153)	Limits ERL differentiation across transaction	Re-run with validated ML probability from real data
Two product types impute (4.7% of dataset)	Mean distribution may not reflect actual reason profile	Collect ST2 output for Clothing Sets and Suits & Sport Coats

Table 20: Limitations of Subtopic 3.

Limitations are summarised in *Table 20*. All cost estimates are sourced from published literature rather than retailer-specific processing records, and the defect cost formula is an approximation that will vary by product type and supplier relationship. The scenario fallback from ST1 produces a narrow probability spread, limiting ERL differentiation across transactions. Two product types required imputation, affecting 4.7% of the dataset. Future work should replace literature costs with retailer-specific values and resolve the imputation for the two missing product types.

Detail	
Input from ST1	p_return_final, order_value_pound, product_type: 85,187 rows
Input from ST2	P(reason product_type) matrix: 24 product types; 2 imputed with mean distribution
Processing	ERL = P(return) x SUM[P(r i) x C(r)]; value-dependent defect cost; Royal Mail shipping tiers; dual ranking; percentile risk tiers

Key output fields	ERL_base, ERL_optimistic, ERL_pessimistic, rank_by_ERL, rank_by_probability, rank_change, risk_tier, top_reason
Output shape	85,187 rows x 22 columns
Output file	ST3_category_summary_for_ST4, ST3_ERL_output_for_ST4

Table 21: ST3 inputs and outputs.

<i>Field</i>	<i>Value</i>
<i>transaction_id</i>	12409
<i>product_category</i>	Swimwear & Active
<i>product_type</i>	Active
<i>order_value_pound</i>	£903.00
<i>p_return_final</i>	0.1297
<i>ERL_base</i>	£11.81
<i>risk_tier</i>	Critical
<i>rank_by_probability</i>	26,271
<i>rank_by_ERL</i>	1
<i>rank_change</i>	+26,270
<i>top_reason</i>	defect

Table 22: Sample output row handed to ST4 (high-ERL transaction).

ST3 produces a transaction-level ERL score, risk tier, and dominant return reason for each of the 85,187 transactions, with inputs and outputs summarized in *Table 21* and a sample high-ERL output row shown in *Table 22*. The ERL score and risk tier determine which transactions warrant intervention; the dominant return reason determines which type of intervention to apply. Sub-Topic 4 uses these outputs to evaluate whether specific interventions are financially justified.

6. Sub-Topic 4: Intervention Evaluation

6.1. Methods

ST4 evaluates whether targeted return-prevention interventions are financially justified using the ST3 ERL scores. The initial design applied twelve transaction-level scenarios to 85,187 transactions matched by top predicted return reason. However, with mean ERL of only £2.20 per transaction, even the cheapest intervention at £0.25 consumed a meaningful share of potential savings before any net benefit could be achieved. A second category-level model was therefore added, pooling ERL exposure by product type and applying strategic pre-transaction interventions at scale. The two approaches are retained because they answer different questions: Approach A identifies rare profitable transaction-level cases; Approach B identifies product groups where intervention is viable at scale.

Approach A: transaction-level intervention model

Approach A defines twelve post-transaction, pre-return interventions across six reason tracks, as detailed in *Table 23*. The model cross-joins every transaction with every scenario and flags only rows where the scenario target reason matches top_reason. Savings are calculated using the relevant reason-specific ERL component where available; delivery scenarios fall back to ERL_base as ST3 contains no delivery-specific cost component.

ID	Scenario	Reason	Reduction	Cost/Tx	Effort	ERL basis
S1	Personalised Fit Outreach	size_fit	7%	£0.25	Low	ERL_from_size_fit
S2	Proactive Size Exchange Offer	size_fit	14%	£0.90	Medium	ERL_from_size_fit
S3	Fast-Track Defect Resolution	defect	12%	£1.20	Medium	ERL_from_defect
S4	Priority Defect Callback	defect	20%	£3.00	High	ERL_from_defect
S5	Post-Purchase Expectation Correction	expectation_mismatch	8%	£0.30	Low	ERL_from_expectation
S6	Personal Styling Follow-Up	expectation_mismatch	15%	£1.50	High	ERL_from_expectation
S7	Delivery Issue Fast-Track Resolution	delivery_issue	10%	£0.70	Medium	ERL_base fallback

S8	Priority Reship for Delivery Failures	delivery_issue	18%	£2.00	High	ERL_base fallback
S9	Retention Offer for Changed-Mind Returns	changed_mind	6%	£0.20	Low	ERL_from_changed
S10	Personalised Retention Outreach	changed_mind	12%	£1.00	Medium	ERL_from_changed
S11	Wrong Item - Fast Reship	wrong_item	15%	£1.50	Medium	ERL_from_wrong_item
S12	Wrong Item - Credit & Keep	wrong_item	20%	£0.80	Low	ERL_from_wrong_item

Table 23: Approach A - transaction-level intervention scenarios.

The recommendation logic produces four outcomes: non-matching rows remain No Action; matched rows with zero or negative net benefit become Monitor; positive rows become Recommend; and rows meeting the stricter priority rule: net benefit $\geq$ £2, payback ratio $\geq$ 2.0, and High, Very High or Critical risk tier, become Prioritise. A priority score combining net benefit, ROI, and ERL magnitude is calculated to rank matched cases, making recommendations interpretable to both analytics and operations users.

Approach B: category-level strategic model

Approach B uses the ST3_category_summary_for_ST4 output from ST3, splitting product types into high- and low-volume groups using the median transaction count of 2,985. Size_fit product types receive either C1a (high-volume size-and-fit overhaul) or C1b (lighter low-volume version). Defect product types receive either C2a quality signaling or C2b lightweight quality content. Category-level net benefits are calculated as: (total ERL $\times$ probability reduction) – (transactions $\times$ cost per transaction). Scenario-level results for the base case are shown in *Table 24*. Product types above £1,000 net benefit are prioritized; positive but smaller opportunities are recommended or monitored; negative cases are marked do not deploy.

ID	Scenario	Target	Types	Tx	Saving	Cost	Net benefit	Avg ROI
C1a	Size & Fit Overhaul - High Volume	size_fit	11	63,428	£32,219	£22,200	+£10,019	44.1%
C1b	Size & Fit Quick Wins - Low Volume	size_fit	9	8,777	£2,418	£1,755	+£662	54.3%

C2b	Quality Signal Lightweight - Low Volume	defect	4	4,624	£869	£694	+£175	28.3%
C2a	Quality Signal Deep Dive - High Volume	defect	2	8,358	£2,255	£2,090	+£166	13.6%

Table 24: Approach B - scenario-level summary, base case.

6.2. Results

Approach A: transaction-level results

Only two return reasons appeared as active targets: size_fit and defect, so S1–S4 produced matches while S5–S12 were retained for future compatibility. Broad rollout produced 170,374 intervention applications, £12,450 estimated saving and £137,560 intervention cost: a net loss of £125,111, as shown in *Table 25*. The high-risk profitable filter reduced this to 31 Critical-tier transactions producing £3.10 net benefit, confirming that ERL can isolate the small subset where transaction-level action is economically justified. The active scenario breakdown under broad rollout is shown in *Table 26*. The broad-rollout loss is a finding, not a flaw: a model that recommends every intervention would be operationally risky and financially weak.

Targeting strategy	Interventions	Saving	Cost	Net benefit
Broad rollout - all matching transactions	170,374	£12,450	£137,560	–£125,111
High-risk profitable only	31	£40.30	£37.20	+£3.10

Table 25: Approach A - broad rollout versus high-risk profitable targeting.

ID	Scenario	Reason	Tx count	Saving	Cost	Net benefit
S1	Personalised Fit Outreach	size_fit	72,205	£2,494	£18,051	–£15,557
S2	Proactive Size Exchange Offer	size_fit	72,205	£4,971	£64,985	–£60,013
S3	Fast-Track Defect Resolution	defect	12,982	£1,869	£15,578	–£13,709
S4	Priority Defect Callback	defect	12,982	£3,115	£38,946	–£35,831

Table 26: Approach A - active scenario breakdown under broad rollout.

Approach B: category-level results

Category-level modelling was substantially stronger. Across 26 product types, the base-case strategy generated £37,760 estimated saving, £26,738 intervention cost, and £11,021 total net benefit. Five product types were classified as Prioritise, six as Recommend, eleven as Monitor, and four as Do not deploy. The value driver was C1a, which alone generated £10,019 net benefit from 63,428 transactions. Jeans produced the highest individual net benefit at

+£2,185, combining the largest transaction volume, high total ERL exposure, and a dominant size_fit reason; full results are in *Table 27*. The category dashboard in *Figure 9* confirms that value is concentrated in Denim & Bottoms and Knitwear & Tops, supporting a staged deployment plan starting with the five Prioritise-tier product types.

Product type	Category	Scenario	Tx	Saving	Cost	Net benefit	ROI	Recommendation
Jeans	Denim & Bottoms	C1a	9,465	£5,498	£3,313	+£2,185	65.9%	Prioritise
Outerwear & Coats	Premium Outerwear	C1a	7,141	£3,726	£2,499	+£1,227	49.1%	Prioritise
Sweaters	Knitwear & Tops	C1a	7,181	£3,635	£2,513	+£1,122	44.6%	Prioritise
Dresses	Occasionwear	C1a	3,369	£2,296	£1,179	+£1,117	94.7%	Prioritise
Swim	Swimwear & Active	C1a	6,990	£3,554	£2,446	+£1,108	45.3%	Prioritise
Pants	Denim & Bottoms	C1a	4,547	£2,494	£1,591	+£902	56.7%	Recommend
Shorts	Denim & Bottoms	C1a	5,362	£2,682	£1,877	+£805	42.9%	Recommend
Fashion Hoodies & Sweatshirts	Knitwear & Tops	C1a	7,392	£3,367	£2,587	+£780	30.2%	Recommend
Suits & Sport Coats	Premium Outerwear	C1a	3,822	£2,001	£1,338	+£663	49.6%	Recommend
Active	Swimwear & Active	C2a	3,570	£1,362	£892	+£469	52.6%	Recommend
Tops & Tees	Knitwear & Tops	C1a	4,212	£1,928	£1,474	+£453	30.7%	Recommend
Pants & Capris	Denim & Bottoms	C2b	1,643	£424	£246	+£177	71.9%	Monitor
Skirts	Denim & Bottoms	C1b	936	£353	£187	+£165	88.3%	Monitor
Blazers & Jackets	Premium Outerwear	C1b	1,712	£482	£342	+£139	40.6%	Monitor
Maternity	Basics & Hosiery	C1b	2,419	£622	£484	+£139	28.6%	Monitor
Leggings	Denim & Bottoms	C1b	502	£186	£100	+£86	85.4%	Monitor
Jumpsuits & Rompers	Occasionwear	C1b	287	£121	£57	+£64	111.1%	Monitor
Suits	Premium Outerwear	C1b	793	£197	£159	+£39	24.3%	Monitor
Clothing Sets	Occasionwear	C1b	159	£69	£32	+£37	116.1%	Monitor
Socks	Basics & Hosiery	C2b	290	£55	£44	+£12	27.3%	Monitor
Plus	Basics & Hosiery	C1b	1,154	£242	£231	+£11	4.7%	Monitor

Socks & Hosiery	Basics & Hosiery	C2b	90	£16	£14	+\$2.45	18.2%	Monitor
Accessories	Accessories	C2b	2,601	£374	£390	-\$17	-4.3%	Do not deploy
Underwear	Intimates & Nightwear	C1b	815	£146	£163	-\$17	-10.2%	Do not deploy
Sleep & Lounge	Intimates & Nightwear	C2a	4,788	£893	£1,197	-\$304	-25.4%	Do not deploy
Intimates	Intimates & Nightwear	C1a	3,947	£1,038	£1,381	-\$343	-24.9%	Do not deploy

Table 27: Approach B - product-type action plan, base case.

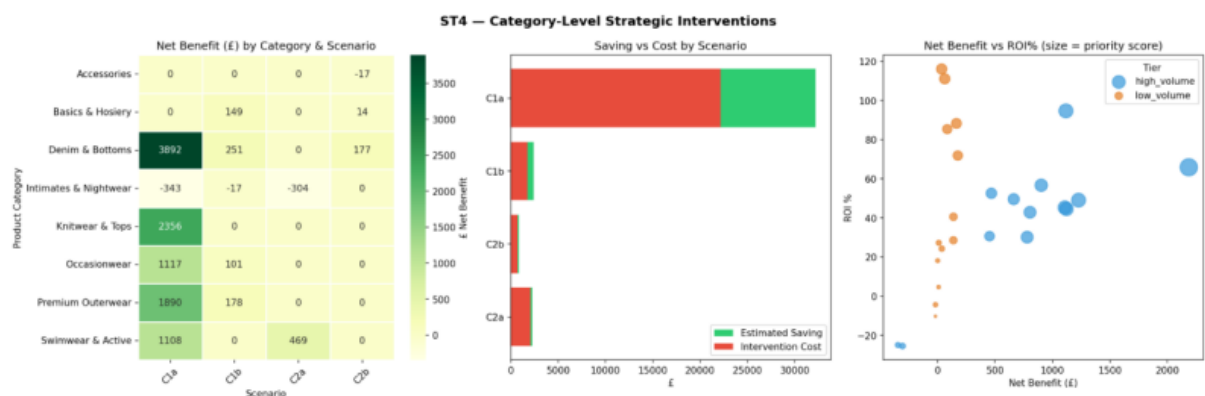


Figure 9: Category-level strategic dashboard: net benefit heatmap, saving-cost comparison and ROI scatter plot.

Sensitivity analysis

As shown in *Table 28*, Approach A remained financially unviable across all scenarios, ranging from -£165,488 in the worst case to -£104,849 in the best case confirming that per-transaction saving is structurally too small relative to intervention cost. Approach B performed substantially better, as shown in *Table 29*: category-level scenarios became profitable at around 75% of the base effectiveness assumption, with C1a moving from -£6,091 at 50% effectiveness to +£26,128 at 150%. Size-and-fit improvements should be prioritised; quality-related interventions should be piloted before full rollout.

Scenario	Worst case (50% eff., ×1.25 cost)	Base case (100% eff.)	Best case (150% eff., ×0.90 cost)	Positive in any case?
S1 — Personalized Fit Outreach	-£21,306	-£15,557	-£12,373	No

S2 — Proactive Size Exchange	−£78,524	−£60,013	−£50,878	No
S3 — Fast-Track Defect Resolution	−£18,534	−£13,709	−£11,219	No
S4 — Priority Defect Callback	−£47,124	−£35,831	−£30,379	No
All scenarios combined	−£165,488	−£125,111	−£104,849	No

Table 28: Sensitivity of Approach A net benefit across worst, base, and best-case assumptions. All values in GBP. Source: ST4 transaction-level intervention model.

Effectiveness assumption	C1a Size & Fit Overhaul (High Vol)	C1b Size & Fit Quick Wins (Low Vol)	C2a Quality Deep Dive (High Vol)	C2b Quality Lightweight (Low Vol)
50% effectiveness (worst case)	−£6,091	−£547	−£962	−£259
75% effectiveness	+£1,964	+£58	−£398	−£42
100% effectiveness (base case)	+£10,019	+£662	+£166	+£175
125% effectiveness	+£18,073	+£1,267	+£729	+£392
150% effectiveness (best case)	+£26,128	+£1,871	+£1,293	+£609

Table 29: Sensitivity of Approach B net benefit (£) across effectiveness assumptions for all four category-level scenarios. Positive values in green; negative values in red. Source: ST4 sensitivity analysis output.

6.3. Discussion and Limitations

The broad-rollout loss in Approach A confirms that at £2.20 mean ERL, per-transaction interventions are financially viable only for the Critical-tier minority that ERL identifies. The shift to Approach B is not a retreat from precision. It is recognition that the correct analytical unit for this dataset's price point is the product group rather than the individual order. Pooling ERL exposure within a product type converts the same per-transaction cost structure from unviable to profitable.

The literature prediction that informational interventions outperform operational ones is confirmed across both approaches: within Approach A, S3 (£1.20; 12% reduction) produces a smaller loss than S4 (£3.00; 20%), and within Approach B the size-and-fit scenarios generate substantially larger net returns than the defect quality-signal scenarios. The recommended deployment sequence is: immediate rollout for the five Prioritise-tier product types; phased expansion to the six Recommend-tier types; no investment in Do Not Deploy types at current ERL levels; and a controlled pilot of higher-ticket transaction targeting before any per-transaction budget commitment.

Three principal limitations qualify these findings, summarised in *Table 30*. First, all probability reduction estimates derive from published benchmarks rather than live trials; the 50% pessimistic scenario turns positive category results negative, so all findings should be treated as a prioritisation framework pending a randomised controlled pilot. Second, *changed_mind* and *wrong_item* carry near-uniform residual probabilities from ST2, making scenarios S9–S12 unreliable; delivery scenarios S7 and S8 fall back to *ERL_base*, which may overstate delivery-related saving. Third, Approach B aggregates within product types and cannot identify high-ERL individual transactions inside low-performing groups; a hybrid model applying transaction-level ERL filtering within product-type cost envelopes would recover this variation.

Limitation	Interpretive Impact	Recommended Improvement
Effectiveness benchmarks from published literature, not live trials	Savings may fall short of modelled projections; 50% worst case turns positive category results negative	Conduct a randomised controlled pilot on a holdout sample before committing budget
<i>changed_mind</i> and <i>wrong_item</i> probabilities are near-uniform residuals from ST2	Scenarios S9–S12 are based on unreliable reason decompositions	Collect dedicated labelled training data for both reason categories
Delivery-issue scenarios fall back to <i>ERL_base</i> (no fulfilment-specific cost component)	<i>ERL_base</i> blends all return cost types and may overstate delivery-specific saving	Extend the ST3 cost matrix with a fulfilment-specific ERL component
Approach B aggregates within product types, masking within-type ERL variation	High-ERL individual transactions inside low-performing types are excluded from targeting	Develop a hybrid model: transaction-level ERL filtering within product-type cost envelopes

Table 30: ST4 limitations, interpretive impact, and recommended improvements

7. Conclusion and Discussion

7.1. Cross-Pipeline Integration and Combined Findings

The main finding of this project is that financial risk and return likelihood are not the same thing. A transaction with a moderate return probability can carry very high financial exposure, and a transaction with a high return probability can be cheap to process. Treating them as equivalent leads to poor prioritisation.

ST1 produced a return probability score for each of the 85,187 transactions. ST2 produced a reason distribution for each product type. ST3 combined both with a cost matrix to produce the ERL score. When cost was introduced alongside probability, 95% of transactions shifted by more than 1,000 ranking positions. A Sleep and Lounge item valued at £399.95 had a moderate return probability of 0.108 but ranked first under ERL because its defect exposure alone reached £95.00. A probability-only system would have placed it near the middle of the dataset.

ST4 extended this finding to the intervention level. The average ERL per transaction was £2.20. At that level, applying interventions broadly to all matching transactions produced a net loss of £125,111. The same ERL scores, grouped by product type, produced £11,021 net benefit across 26 product types. This result required all four sub-topic outputs working in sequence. It could not have been reached from any single stage alone. The Gini coefficient improvement from 0.1058 to 0.1601 is also a joint result. It depends on ST2's reason distributions separating cost exposure by product type, and on ST3's value-dependent defect cost amplifying that separation for high-value transactions.

Limitations in the earlier stages affected all stages that followed. ST1 did not reach the AUC-ROC threshold of 0.80. The scenario-based fallback produced a narrow probability spread of 0.065 to 0.153. This constrained ERL differentiation in ST3 and kept the average ERL low enough that transaction-level intervention was rarely viable in ST4. ST2 could not learn reliable predictions for `changed_mind` and `wrong_item`. These two categories produced near-uniform outputs that added noise to ST3's cost calculation and made scenarios S9 to S12 in ST4 unreliable.

Both limitations point in the same direction. The 51% Gini improvement and £11,021 category-level net benefit were achieved under the least favourable input conditions the framework is likely to face. With validated ML probabilities and dedicated labelled data for the two residual categories, every downstream result would be expected to improve. A probability spread of 0.05 to 0.60, consistent with the deployment validation run in Appendix

A, would raise the mean ERL from £2.20 to approximately £8 to £12 per transaction. At that level, the cheapest intervention would be profitable for most Critical-tier transactions rather than only 31.

7.2. Overall Conclusion

This project set out to build a single financial risk score for e-commerce returns by combining three things no published framework had combined before: return probability, return reason, and processing cost. That score now exists, runs across 85,187 transactions, and produces rankings that are meaningfully different from anything a probability-only system would generate.

The results show clearly that financial risk and return likelihood are not the same thing. A transaction can be likely to be returned and cheap to process. Another can be unlikely to be returned and financially devastating if it is. Treating these as equivalent leads to the wrong decisions. The ERL framework separates them.

Every component was validated or architecturally confirmed. ST1 confirmed the pipeline code is correct through a deployment run on alternative data. ST2 reliably classified the three reason categories that drive the majority of return costs. ST3 demonstrated that introducing cost into the ranking changes where attention goes. ST4 showed that targeting by financial exposure outperforms targeting by probability, and identified exactly where intervention is and is not worth the cost.

The results were achieved on synthetic data with literature-based cost estimates. That is the honest limitation. It is also the clearest possible signal of what comes next. The framework does not need to be rebuilt. It needs to be fed real data: real return rates, real processing costs, and real intervention outcomes from a controlled pilot. When that happens, every metric in this report has room to improve.

8. Author Contributions

Author contributions are described using the CRediT taxonomy (Brand et al., 2015).

Dang Nguyet Anh Nguyen: Methodology; Software (ST1); Formal analysis (ST1); Writing, original draft (ST1); Writing, review and editing.

Sawsan Hussain Saleh Ibrahim: Methodology; Software (ST2); Data curation; Formal analysis (ST2); Writing, original draft (ST2); Writing, review and editing.

Julia Legner: Methodology; Software (ST3); Formal analysis (ST3); Visualisation; Writing, original draft (ST3); Writing, review and editing.

Afrida Nawer Prami: Methodology; Software (ST4); Formal analysis (ST4); Investigation; Writing, original draft (ST4); Writing, review and editing.

All authors contributed to Writing, review and editing, and approved the final manuscript.

9. References

- Alfonso, V., Boar, C., Frost, J., Gambacorta, L. and Liu, J. (2021) E-commerce in the pandemic and beyond. *BIS Bulletin*, 36, pp.1–9. Available at: https://www.bis.org/publ/bisbull36_appendix.pdf [Accessed 24 May 2026].
- Barclaycard (2016) *The Habit of Returns*. Available at: <https://re-turns.com/research-from-barclaycard/> [Accessed 25 May 2026].
- Basel Committee on Banking Supervision (2004) *Basel II: International Convergence of Capital Measurement and Capital Standards: A Revised Framework*. Bank for International Settlements. Available at: <https://www.bis.org/publ/bcbs107.htm> [Accessed 24 May 2026].
- BCS (2022) *BCS Code of Conduct*. Available at: <https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/> [Accessed 25 May 2026].
- Cui, H., Rajagopalan, S. and Ward, A.R. (2020) Predicting product return volume using machine learning methods. *European Journal of Operational Research*, 281(3), pp.612–627. Available at: <https://doi.org/10.1016/j.ejor.2019.05.046> [Accessed 24 May 2026].
- Devlin, J., Chang, M.W., Lee, K. and Toutanova, K. (2019) BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, pp.4171–4186. Available at: <https://arxiv.org/abs/1810.04805> [Accessed 25 May 2026].
- Elkan, C. (2001) The foundations of cost-sensitive learning. *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI 2001)*, pp.973–978. Available at: <https://cseweb.ucsd.edu/~elkan/rescale.pdf> [Accessed 24 May 2026].
- Hinton, G., Vinyals, O. and Dean, J. (2015) Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*. Available at: <https://arxiv.org/abs/1503.02531> [Accessed 25 May 2026].
- Hosmer, D.W., Lemeshow, S. and Sturdivant, R.X. (2013) *Applied Logistic Regression*. 3rd edn. Hoboken: Wiley. Available at: <https://doi.org/10.1002/9781118548387> [Accessed 25 May 2026].
- IMRG (2022) *Returns in Fashion Ecommerce*. London: IMRG.
- Karl, D. (2025) Forecasting e-commerce consumer returns: a systematic literature review. *Management Review Quarterly*, 75(3), pp.1–56. Available at: <https://doi.org/10.1007/s11301-024-00436-x> [Accessed 25 May 2026].

- Keeney, R.L. and Raiffa, H. (1976) *Decisions with Multiple Objectives: Preferences and Value Trade-offs*. New York: Wiley.
- Landis, J.R. and Koch, G.G. (1977) The measurement of observer agreement for categorical data. *Biometrics*, 33(1), pp.159–174. Available at: <https://doi.org/10.2307/2529310> [Accessed 25 May 2026].
- Malik, N. and Bilal, M. (2024) NLP for analysing online customer reviews using machine learning approaches. *PeerJ Computer Science*, 10, e2203. Available at: <https://doi.org/10.7717/peerj-cs.2203> [Accessed 25 May 2026].
- Mishra, A. and Dutta, P. (2024) Return management in e-commerce firms: a sustainable perspective. *Journal of Cleaner Production*, 477, 143802. Available at: <https://doi.org/10.1016/j.jclepro.2024.143802> [Accessed 25 May 2026].
- Narvar (2019) *Making Returns a Competitive Advantage*. Narvar Inc.
- National Retail Federation and Happy Returns (2024) *2024 Consumer Returns in the Retail Industry*. Washington DC: National Retail Federation. Available at: https://a-us.storyblok.com/f/1021220/x/84c63128d9/2024-consumer-returns-in-the-retail-industry-report_12-5-24-2.pdf [Accessed 24 May 2026].
- Nestler, A., Hadder, M. and Radev, I. (2021) SizeFlags: detecting size and fit related customer returns in fashion e-commerce. *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2021)*, pp.3430–3438. Available at: <https://doi.org/10.1145/3447548.3467160> [Accessed 25 May 2026].
- Ni, J., Li, J. and McAuley, J. (2019) Justifying recommendations using distantly-labelled reviews and fine-grained aspects. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP 2019)*, pp.188–197. Available at: <https://aclanthology.org/D19-1018/> [Accessed 25 May 2026].
- Niculescu-Mizil, A. and Caruana, R. (2005) Predicting good probabilities with supervised learning. *Proceedings of the 22nd International Conference on Machine Learning (ICML 2005)*, pp.625–632. Available at: <https://dl.acm.org/doi/10.1145/1102351.1102430> [Accessed 25 May 2026].
- Petersen, J.A. and Kumar, V. (2009) Are product returns a necessary evil? Antecedents and consequences. *Journal of Marketing*, 73(3), pp.35–51. Available at: <https://doi.org/10.1509/jmkg.73.3.35> [Accessed 25 May 2026].
- Platt, J. (2000) Probabilistic outputs for support vector machines and comparisons to regularised likelihood methods. In Smola, A., Bartlett, P., Schölkopf, B. and Schuurmans, D. (eds.) *Advances in Large Margin Classifiers*. Cambridge: MIT Press, pp.61–74.

- Ratner, A., De Sa, C., Wu, S., Selsam, D. and Ré, C. (2017) Data programming: creating large training sets quickly. *Proceedings of the VLDB Endowment*, 11(3), pp.269–282. Available at: <https://doi.org/10.14778/3157794.3157797> [Accessed 25 May 2026].
- Ren, S., Choi, T.M. and Liu, N. (2021) Fashion sales forecasting with a panel data-based particle-filter model. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 45(3), pp.411–421. Available at: <https://doi.org/10.1109/TSMC.2014.2364681> [Accessed 25 May 2026].
- Roichman, R., Sprecher, B., Blass, V., Meshulam, T. and Makov, T. (2024) The convenience economy: product flows and GHG emissions of returned apparel in the EU. *Resources, Conservation and Recycling*, 219, 107811. Available at: <https://doi.org/10.1016/j.resconrec.2024.107811> [Accessed 25 May 2026].
- Royal Mail (2024) *Tracked Returns: Business Price Guide*. London: Royal Mail Group Ltd. Available at: <https://www.royalmail.com/sending/returns/tracked-returns> [Accessed 25 May 2026].
- Saito, T. and Rehmsmeier, M. (2015) The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. Available at: <https://doi.org/10.1371/journal.pone.0118432> [Accessed 25 May 2026].
- Sanh, V., Debut, L., Chaumond, J. and Wolf, T. (2019) DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*. Available at: <https://arxiv.org/abs/1910.01108> [Accessed 25 May 2026].
- Tabassi, E. (2023) *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. Gaithersburg: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.AI.100-1> [Accessed 25 May 2026].
- Zadrozny, B. and Elkan, C. (2002) Transforming classifier scores into accurate multiclass probability estimates. *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2002)*, pp.694–699. Available at: <https://dl.acm.org/doi/10.1145/775047.775151> [Accessed 25 May 2026].

10. Appendix

Appendix A: ST1 Deployment Validation Run

A1 — Category Return-Rate Diagnostic

Table A1: Observed return rate by product category on the alternative synthetic dataset.

Product Category	Return Rate	Returns (n)	Total (n)
Knitwear & Tops	0.5985	11,243	18,785
Accessories	0.4910	1,277	2,601
Swimwear & Active	0.4098	4,328	10,560
Basics & Hosiery	0.3322	1,313	3,953
Premium Outerwear	0.2522	3,397	13,468
Occasionwear	0.1777	678	3,815
Intimates & Nightwear	0.0990	945	9,550
Denim & Bottoms	0.0188	423	22,455

Table A1 shows the observed return rate per product category on the alternative synthetic dataset. The cross-category standard deviation is **0.1983**, compared with **0.0036** in the primary dataset. This wide spread confirms that the features available at purchase time carry sufficient information to distinguish high-risk from low-risk transactions - the condition under which a trained classifier can produce discriminative output.

A2 — Validation Model Comparison and Threshold Decision

Table A2: Validation model comparison on the alternative synthetic dataset

Model	AUC-ROC	Avg Precision	Brier	Meets Threshold
Random Forest	0.8070	0.5422	0.1536	Yes
Logistic Regression	0.8016	0.5317	0.1548	Yes

Both candidate classifiers were evaluated on the validation partition (*Table A2*). Random Forest achieved a validation AUC-ROC of **0.8070**, satisfying the pre-defined threshold of $\geq$

0.80 and triggering the calibrated machine-learning handoff path. The pipeline therefore did not fall back to the scenario score, as it did in the primary dataset run.

A3 — Final Test Assessment

Table A3: Final test assessment on the alternative synthetic dataset.

Model / Benchmark	AUC-ROC	Avg Precision	Brier	Meets Threshold
Random Forest (test)	0.8091	0.5533	0.1536	Yes
Base-rate baseline	0.5000	0.2789	0.2011	No
p_return (diagnostic)	0.4955	0.2771	0.2224	No

Once the model was selected on the validation partition, it was evaluated once on the held-out test partition. Random Forest achieved a test AUC-ROC of **0.8091** (Table A3), confirming that validation performance was not an artefact of partition choice.

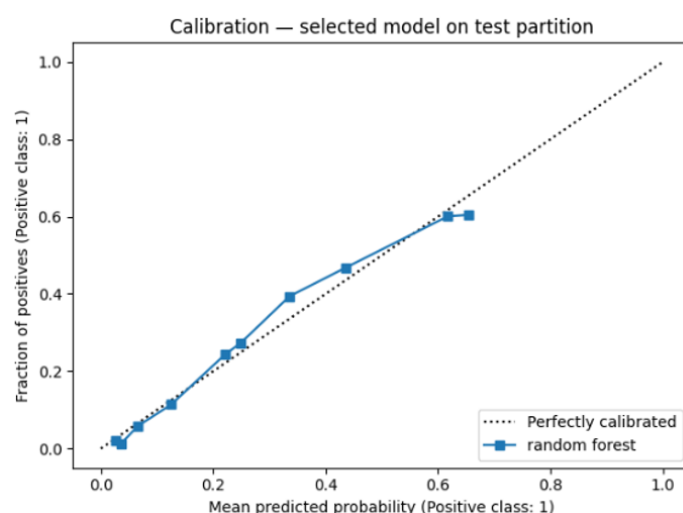


Figure A1: Calibration of the selected model (Random Forest) on the test partition - alternative synthetic dataset.

Calibration on the test partition is confirmed in Figure A1. The blue line closely tracks the perfectly calibrated reference, confirming that predicted probabilities correspond to observed return rates across the full score range.

Appendix B: ST2: Return Reason Classification Pipeline and

Full P(reason | product_type) Matrix

Overview

Goal: Classify each of the 33,835 TheLook transactions into one of five return reason categories, then aggregate to P(reason | product_category) for ST3.

Input: TheLook 33,835 rows + 124,979 Amazon Apparel reviews (star_rating <= 2). Output: table10_p_reason_product_type.csv

ID	Label	Meaning
0	size_fit	Wrong size or did not fit
1	defect	Damaged or poor quality
2	expectation_mismatch	Did not match description or photos
3	wrong_item	Received a different product
4	changed_mind	Customer no longer wanted it

End-to-end pipeline

Stage	Input	Process	Output
1. Preprocess	Amazon TSV + TheLook 33k	Concat headline+body, decode HTML, filter star <= 2	Clean review text
2. 23 LFs	Review text + star_rating	Phrase-based + co-occurrence rules vote 0-4 or abstain (-1)	Label matrix L (n x 23)
3. LabelModel	Label matrix L	Learn LF accuracy + correlations via EM, apply Bayes	Soft 5-dim probability per row
4. Confidence filter	Soft probabilities	Keep rows where max(prob) >= 0.65	270,842 high-confidence rows
5. DistilBERT	20k stratified sample	Fine-tune on soft labels using KL divergence loss	Fine-tuned model, macro F1 = 0.737
6. Inference	All 33,835 TheLook rows	Model inference then average by product_type	table10_p_reason_product_type.csv

Sample data through the pipeline (same 3 reviews)

Vote key: -1 = abstain | 0 = size_fit | 1 = defect | 2 = expectation_mismatch | 3 = wrong_item | 4 = changed_mind

Review text	star	lf_def_explicit	lf_def_cooccu	lf_size_explicit	lf_size_cooccu	lf_expectation	lf_return_confir
zipper broke after one wash, sizing runs two sizes small	1	1=defect	1=defect	-1	-1	-1	1=defect
arrived different from the photo, wrong colour	2	-1	-1	-1	-1	2=expect	-1
ordered medium but fits like XS, totally unwearable	1	-1	-1	0=size	0=size	-1	-1

Label matrix L: full 23-column shape (3 columns shown for brevity):

row	lf_01	lf_02	...	lf_23	Non-abstain votes	Dominant votes
0	1	1	...	-1	3 x defect	defect
1	-1	-1	...	-1	1 x expect	expectation_mismatch
2	0	-1	...	0	2 x size_fit	size_fit

LabelModel output: soft probabilities (LF accuracy + correlation discounting + class prior applied):

row	p_size_fit	p_defect	p_expect_m	p_wrong_item	p_changed_min	top_reason	max_prob	kept?
0	0.06	0.81*	0.04	0.03	0.06	defect	0.81	YES
1	0.10	0.08	0.66*	0.11	0.05	expectation_mismatch	0.66	YES
2	0.72*	0.08	0.14	0.03	0.03	size_fit	0.72	YES
3	0.38	0.16	0.24	0.14	0.08	size_fit	0.38	NO

* dominant probability. 270,842 rows pass the confidence filter. Rows with all-abstain LFs output the class prior e.g. [0.38, 0.29, 0.24, 0.05, 0.04] and are typically dropped.

Phase 3: Manual annotation validation

Metric	Value	Threshold	Result
Cohen's kappa (inter-annotator)	0.91	-	Almost perfect agreement
Snorkel vs. human agreement	82.4%	70%	PASSED

Phase 4: DistilBERT fine-tuning

Trained on Snorkel soft labels (not hard classes) using KL divergence loss: preserves uncertainty across all five classes. 20k stratified sample, 60/20/20 split, AdamW lr=2e-5, early stopping on validation loss.

Why KL divergence not cross-entropy: cross-entropy collapses [0.81, 0.06, 0.04, 0.03, 0.06] to defect=1, rest=0: discarding the uncertainty. KL divergence trains the model to match the full distribution.

Category	F1	Target >= 0.68?	Category	F1	Target >= 0.68?
size_fit	0.965	YES	wrong_item	0.869	YES
defect	0.970	YES	changed_mind	0.000	No: absent from low-star reviews
expect_mm	0.881	YES	Macro average	0.737	YES

Final output: P(reason | product_category)

DistilBERT runs inference on all 33,835 TheLook rows. Probability vectors averaged within each product_type group. Output feeds ST3 ERL formula as the P(r|i) term.

product_category	p_size_fit	p_defect	p_expect_mm	p_wrong_item	p_changed_mind
Jeans	0.61	0.12	0.18	0.05	0.04
Dresses	0.38	0.09	0.41	0.07	0.05
Tops	0.45	0.15	0.28	0.06	0.06
...	...	...	...	...	...

Final output: Full P(reason | product_type) Matrix

DistilBERT runs inference on all 33,835 TheLook rows. Probability vectors averaged within each product_type group. This is table10_p_reason_product_type.csv — the P(r|i) input to ST3. Note: wrong_item and changed_mind are softmax residuals, not genuine predictions (F1 = 0 for changed_mind).

product_type	p_size_fit	p_defect	p_expect_mm	p_wrong_item	p_changed_mind
Accessories	0.2707	0.4276	0.1262	0.0712	0.1043
Active	0.2744	0.4279	0.1231	0.0697	0.1048
Blazers & Jackets	0.6295	0.2183	0.0619	0.0310	0.0593
Dresses	0.4179	0.3241	0.1253	0.0495	0.0832
Fashion Hoodies & Sweatshirts	0.3861	0.3334	0.1308	0.0557	0.0940
Intimates	0.4155	0.3266	0.1025	0.0571	0.0983
Jeans	0.4369	0.3054	0.1085	0.0536	0.0956
Jumpsuits & Rompers	0.5192	0.3224	0.0522	0.0481	0.0582
Leggings	0.3964	0.3529	0.1163	0.0499	0.0846
Maternity	0.4627	0.3010	0.0942	0.0497	0.0925
Outerwear & Coats	0.4524	0.3102	0.1012	0.0494	0.0868
Pants	0.3786	0.3577	0.1145	0.0567	0.0926
Pants & Capris	0.3610	0.3721	0.1016	0.0623	0.1030
Plus	0.6346	0.1244	0.0986	0.0539	0.0884
Shorts	0.4365	0.3092	0.1108	0.0524	0.0912
Skirts	0.4268	0.3243	0.1191	0.0492	0.0805
Sleep & Lounge	0.3479	0.3814	0.1112	0.0615	0.0980
Socks	0.3394	0.3738	0.1207	0.0643	0.1018
Socks & Hosiery	0.3552	0.3597	0.1452	0.0544	0.0855
Suits	0.5671	0.1454	0.0984	0.0628	0.1263
Sweaters	0.3806	0.3406	0.1323	0.0560	0.0906

Swim	0.3931	0.3439	0.1182	0.0544	0.0904
Tops & Tees	0.3864	0.3420	0.1150	0.0593	0.0973
Underwear	0.3717	0.3602	0.1033	0.0630	0.1017

Dominant reason per type: size_fit leads in fit-sensitive categories (Plus 0.635, Blazers 0.630, Jumpsuits 0.519). Defect leads in accessories and active wear (Active 0.428, Accessories 0.428). All rows sum to 1.0 via softmax.

Appendix C: Code and Reproducibility

The full codebase is submitted as a zip file (Group2_COMP1884.zip) accompanying this report. All notebooks run in Google Colab (Python 3). Pre-computed outputs are included at every stage so nothing needs to be re-run to verify results.

The pipeline runs sequentially across four notebooks: ST1/st1_main_pipeline.ipynb, then ST2/ST2_Submission/st2_pipeline.ipynb, then ST3/st3_pipeline.ipynb, then ST4/st4_pipeline_v1.ipynb. A supplementary category-level analysis is in ST4/ST4_category_level_v2.ipynb. Full run instructions are in the README.md at the root of the zip and in each sub-topic folder.

To verify results without running any code, ST3 full outputs are in ST3/st3_pipeline_colab.pdf and ST4 scenario results are in ST4/outputs/ST4_intervention_summary.xlsx.

The Amazon Apparel review corpus (amazon_reviews_us_Apparel_v1_00.tsv, approximately 4 GB) is not included in the zip due to file size. The derived dataset used for DistilBERT training (combined_high_confidence.csv, 270k rows) is included. A 5k sample is included for end-to-end testing of the Snorkel pipeline.

Appendix D: ST3 Risk Tier Reason Decomposition

Table D1 extends the headline risk tier results from Table 20 by showing the dominant return reason breakdown within each tier. As defect cost is value-dependent, higher-value transactions concentrate in the upper tiers, explaining the increasing defect share as ERL rises — from 52% in the Low tier to 75% in the Critical tier. The near-uniform return rates across tiers (11.6% to 12.2%, reported in Table 20) confirm that tier assignment reflects cost concentration rather than return frequency.

Risk Tier	Transactions	Avg ERL	% Defect	% Size-fit	% Expect. mismatch
Low	42,675	£1.74	52%	25%	12%
Medium	21,528	£2.28	55%	22%	12%
High	12,589	£2.66	61%	20%	10%
Very High	4,250	£3.14	66%	17%	8%
Critical	4,145	£4.20	75%	13%	6%

Table D1: Risk tier breakdown by dominant return reason. Percentages reflect the proportion of total ERL exposure within each tier attributable to each reason category. *Wrong_item* and *changed_mind* account for the remaining share (approximately 11% in the Low tier, declining to 6% in the Critical tier) and are omitted from the table for brevity.

Appendix E: ERL Formula Worked Examples– ST3

The following two examples demonstrate the full ERL calculation for one Critical-tier and one Low-tier transaction, confirming that the formula is applied consistently and that outputs are interpretable at transaction level.

Example 1 — Critical tier: Transaction 12409 (Swimwear & Active, £903.00)

- P(return): 0.1297
- Defect cost: £15.00 + (20% × £903.00) = **£195.60**
- P(reason) × C(reason):
 - size_fit: 0.2744 × £10.00 = £2.74
 - defect: 0.4279 × £195.60 = £83.70
 - expectation_mismatch: 0.1231 × £18.00 = £2.22
 - wrong_item: 0.0697 × £22.00 = £1.53
 - changed_mind: 0.1048 × £8.00 = £0.84
- **Weighted processing cost: £91.03**
- **ERL = 0.1297 × £91.03 = £11.81**

Example 2 — Low tier: representative low-value transaction (Accessories, £35.00)

- P(return): 0.064 (scenario score for Accessories category)
- Reason probabilities from ST2 (Accessories product type):
- P(reason) × C(reason):
 - size_fit: $0.271 \times £10.00 = £2.71$
 - defect: $0.428 \times £22.00^* = £9.42$
 - expectation_mismatch: $0.126 \times £18.00 = £2.27$
 - wrong_item: $0.071 \times £22.00 = £1.56$
 - changed_mind: $0.104 \times £8.00 = £0.83$
- *Defect cost = $£15.00 + (20\% \times £35.00) = £22.00$
- Weighted processing cost: £16.79
- **ERL = $0.064 \times £16.79 = £1.07$**

This transaction sits in the Low tier with an ERL of £1.07, below the average of £1.74 for that tier. Despite Accessories having the highest defect probability of any product type (0.428), the low order value keeps the defect cost modest, confirming that both the reason distribution and the order value jointly determine ERL.

Appendix F: ST4 Worked Examples

This appendix shows one worked example for each ST4 approach. The purpose is to make the calculation behind the ST4 net benefit results easier to check.

F1 — Approach A: Transaction-Level Example

This example uses transaction 12409, the same high-value swimwear item used in Appendix E. Because its dominant return reason is defect, it is matched to Scenario S3: Fast-Track Defect Resolution.

Step	Calculation	Result
1. Defect-related ERL	$0.1297 \times 0.4279 \times £195.60$	≈ £10.86
2. Estimated saving (12% reduction)	$£10.86 \times 0.12$	≈ £1.30
3. Intervention cost (S3)	Fixed cost per transaction	£1.20
4. Net benefit	$£1.30 - £1.20$	≈ £0.10

Table F1: Step-by-step calculation for transaction 12409 under Scenario S3.

The calculation first estimates the defect-related expected loss for this transaction. It then applies the 12% reduction assumed for Scenario S3 and subtracts the fixed intervention cost. The result is a small positive net benefit of about £0.10 for this single transaction.

This same calculation is repeated across all matched transactions. Only the profitable Critical-tier cases are kept for the high-risk profitable filter, while broad rollout includes many negative cases, producing an overall loss.

F2 — Approach B: Category-Level Example

This example uses Jeans under Scenario C1a: Size & Fit Overhaul – High Volume.

Step	Calculation	Result
1. Total estimated saving	Total size_fit ERL across all Jeans transactions × probability reduction	£5,497.50
2. Total intervention cost	9,465 transactions × cost per transaction	£3,312.75
3. Net benefit	£5,497.50 – £3,312.75	+£2,184.75
4. ROI	$(£2,184.75 / £3,312.75) \times 100$	≈ 65.9%

Table F2: Step-by-step calculation for Jeans under Scenario C1a (9,465 transactions).

Approach B works at product-type level rather than individual transaction level. It first sums the size-fit ERL across all Jeans transactions, then applies the assumed probability reduction and subtracts the total intervention cost. The result is a positive net benefit of £2,184.75 and an ROI of about 65.9%.

This shows why category-level targeting performs better than broad transaction-level rollout. Individual ERL values are often too small to justify intervention costs, but when grouped across thousands of transactions, the savings become large enough to produce a positive result.

11. Declaration of AI Use

We have used AI while undertaking our assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources: NO